GS/ OIEC/ IOEC	HYDRAULIC DATA SHEETS	N.I.O.C Pars Oil & Gas Company

EQUIPMENT NAME: **COLUMN INTERNALS FOR CONDENSATE STABILISER**

DOCUMENT TITLE: **HYDRAULIC DATA SHEETS**



F	Jun.19 '06	FINAL	D.S.OHK	B.D.LEE	E.Y.LEE
1	Apr.27 '05	REVISED BY COMMENTS	D.S.OHK	B.D.LEE	E.Y.LEE
0	Mar.07 '05	FOR APPROVAL	D.S.OHK	B.D.LEE	E.Y.LEE
REV.	DATE	DESCRIPTION	PRE'D	CHK'D	APP'D

SOUTH PARS GAS FIELD DEVELOPMENT (PHASES 9&10), ASSALUYEH, IRAN ONSHORE FACILITIES	CONTRACT NO.: POGC-760-81-156
	REQ'N NO.: DI-6340S-0510-LG-0040
HMT	ITEM NO.: 103-C-101/201
	DOCUMENT NO.: VP-6340S-103-CI101-001

VENDOR NAME: HANBAL MASSTECH LIMITED

HANBAL VALVE TRAY DESIGN PROGRAM VERSION 1.0.0T LAYOUT DIMENSIONS 15-Dec-2004 10:55 AM

CLIENT: NATIONAL IRANIAN OIL COMPANY
COLUMN: 103-C-101/CONDENSATE STABILIZER
CASE:

DESIGNER: D.S.OHK
REVISED: 0

NUMBER OF PASSES	2		
DIAMETER-MM:	2600.0000		
SLOPED DOWNCOMERS			
CHORD HEIGHTS-MM	TOP	BOTTOM	
SIDE	450.000	385.000	
O/C OUTSIDE	0.000	0.000	
O/C INSIDE	0.000	0.000	
CENTER	1065.000	1180.000	
D/C AREAS-SQ.M			RATIO-BTM/TOP
SIDE	0.614	0.490	0.7980
OFF-CENTER	0.000	0.000	0.0000
CENTER	1.215	0.623	0.5127
BUBBLING AREA-SQ.M	3.1137		
EXIT WEIR LENGTHS-MM			
SIDE	1967.232	1846.916	
O/C OUTSIDE	0.000	0.000	
O/C INSIDE	0.000	0.000	
CENTER	2557.166	2588.899	
FLOWPATH WIDTH-MM	4569.4781		
FLOWPATH LENGTH-MM	680.0000	(MINIMUM LENGTH)	

TRAY NO.	2 - 16	2 - 16	2 - 16
HEADINGS	#9(110%)	#9(100%)	#9(30%)
TRAY SPACING-MM	600.000	600.000	600.000
UNDERDOWNFLOW-MM	75.000	75.000	75.000
INLET WEIR HEIGHT-MM	0.000	0.000	0.000
EXIT WEIR HEIGHT-MM	80.000	80.000	80.000
PERCENT BLOCK	0.0	0.0	0.0
VALVES PER TRAY	394	394	394
LIGHT VALVES PER TRAY	138	138	138
PERCENT HOLE AREA	15.1	15.1	15.1

NOTES:

DESIGN FOR QUOTE/(ORDER) APPROVED BY E. Y. Lee

HANBAL VALVE TRAY DESIGN PROGRAM VERSION 1.0.0T PROCESS DATA 15-Dec-2004 10:55 AM

CLIENT: NATIONAL IRANIAN OIL COMPANY
COLUMN: 103-C-101/CONDENSATE STABILIZER
CASE: SUMMER CASE

DESIGNER: D.S.OHK
REVISED: 0

TRAY NO.	2 - 16	2 - 16	2 - 16
HEADINGS	#9(110%)	#9(100%)	#9(30%)
VAPOR FLOW-KGS/HR	76689.8	69718.0	20915.4
VLOAD-M3/HR	669.70	608.82	182.65
DENSITY-KGS/M3	22.71326	22.71326	22.71326
LIQUID FLOW-KGS/HR	256793.9	233449.0	70034.7
FLOW-M3/HR	427.95	389.04	116.71
DENSITY-KGS/M3	600.055	600.055	600.055
SYSTEM FACTOR-VAPOR	1.000	1.000	1.000
SYSTEM FACTOR-LIQUID	1.000	1.000	1.000
JET FLOOD PERCENT	65.0	59.1	17.8
D/C BACKUP FLOOD PERCENT	64.8	58.9	17.7
D/C LOAD PERCENT SAFE DESIGN	82.4	74.9	22.5
WEEP POINT PERCENT OF DESIGN	24.0	26.3	85.6
P.D./TRAY-KG/CM2	0.006361	0.005995	0.003895
TRAY P.D.-MM.LIQ.	106.004	99.901	64.917
DRY TRAY P.D.-MM.LIQ.	32.813	28.440	9.494
TRAY FOAM HEIGHT-MM.LIQ.	126.050	124.281	88.233
UDFC P.D.-MM.LIQ.	30.062	24.845	2.236
D/C BACKUP-MM.LIQ.	262.116	249.027	155.386
-MM FOAM	407.911	371.101	159.746
FH HOLE VELOCITY FACTOR-M/HR	1422.694	1293.358	388.008
FH RECOMMENDED SAFE DESIGN-M/HR	2387.306	2387.306	2387.306
D/C LOADS-M3/M2/HR			
SIDE	348.4	316.7	95.0
OFF-CENTER	0.0	0.0	0.0
CENTER	352.1	320.1	96.0
WEIR LOADS-M3/M/HR			
EFFECTIVE	108.8	98.9	29.7
SUPERFICIAL			
SIDE	108.8	98.9	29.7
O/C OUTSIDE	0.0	0.0	0.0
O/C INSIDE	0.0	0.0	0.0
CENTER	83.7	76.1	22.8
HEIGHT OVER EXIT WEIR-MM.LIQ	64.539	60.563	27.130
SIDE UDF VELOCITY-M/HR	1544.730	1404.300	421.290

HANBAL VALVE TRAY DESIGN PROGRAM VERSION 1.0.0T PROCESS DATA 15-Dec-2004 10:55 AM

CLIENT: NATIONAL IRANIAN OIL COMPANY
COLUMN: 103-C-101/CONDENSATE STABILIZER
CASE: WINTER CASE

DESIGNER: D.S.OHK
REVISED: 0

TRAY NO. / HEADINGS	2 - 16 #9(110%)	2 - 16 #9(100%)	2 - 16 #9(30%)
VAPOR FLOW-KGS/HR	70737.7	64307.0	19292.1
VLOAD-M3/HR	625.86	568.96	170.69
DENSITY-KGS/M3	21.94526	21.94526	21.94526
LIQUID FLOW-KGS/HR	260630.7	236937.0	71081.1
FLOW-M3/HR	431.46	392.24	117.67
DENSITY-KGS/M3	604.062	604.062	604.062
SYSTEM FACTOR-VAPOR	1.000	1.000	1.000
SYSTEM FACTOR-LIQUID	1.000	1.000	1.000
JET FLOOD PERCENT	62.0	56.4	16.9
D/C BACKUP FLOOD PERCENT	64.2	58.4	17.5
D/C LOAD PERCENT SAFE DESIGN	83.1	75.5	22.7
WEEP POINT PERCENT OF DESIGN	25.9	28.3	92.5
P.D./TRAY-KG/CM2	0.006220	0.005884	0.003912
TRAY P.D.-MM.LIQ.	102.966	97.410	64.759
DRY TRAY P.D.-MM.LIQ.	29.622	25.803	9.257
TRAY FOAM HEIGHT-MM.LIQ.	126.173	124.455	88.446
UDFC P.D.-MM.LIQ.	30.557	25.254	2.273
D/C BACKUP-MM.LIQ.	259.697	247.118	155.478
-MM FOAM	405.694	369.595	160.093
FH HOLE VELOCITY FACTOR-M/HR	1329.553	1208.685	362.605
FH RECOMMENDED SAFE DESIGN-M/HR	2387.306	2387.306	2387.306
D/C LOADS-M3/M2/HR			
SIDE	351.2	319.3	95.8
OFF-CENTER	0.0	0.0	0.0
CENTER	355.0	322.7	96.8
WEIR LOADS-M3/M/HR			
EFFECTIVE	109.7	99.7	29.9
SUPERFICIAL			
SIDE	109.7	99.7	29.9
O/C OUTSIDE	0.0	0.0	0.0
O/C INSIDE	0.0	0.0	0.0
CENTER	84.4	76.7	23.0
HEIGHT OVER EXIT WEIR-MM.LIQ	64.892	60.895	27.278
SIDE UDF VELOCITY-M/HR	1557.410	1415.827	424.748

HANBAL VALVE TRAY DESIGN PROGRAM VERSION 1.0.0T LAYOUT DIMENSIONS 15-Dec-2004 10:55 AM

CLIENT: NATIONAL IRANIAN OIL COMPANY
COLUMN: 103-C-101/CONDENSATE STABILIZER
CASE:

DESIGNER: D.S.OHK
REVISED: 0

NUMBER OF PASSES	1		
DIAMETER-MM:	2600.0000		
STRAIGHT DOWNCOMER			
CHORD HEIGHTS-MM	TOP	BOTTOM	
SIDE	260.000	260.000	
O/C OUTSIDE	0.000	0.000	
O/C INSIDE	0.000	0.000	
CENTER	0.000	0.000	
D/C AREAS-SQ.M			RATIO-BTM/TOP
SIDE	0.276	0.276	1.0000
OFF-CENTER	0.000	0.000	0.0000
CENTER	0.000	0.000	0.0000
BUBBLING AREA-SQ.M	4.7567		
EXIT WEIR LENGTHS-MM			
SIDE	1560.000	1560.000	
O/C OUTSIDE	0.000	0.000	
O/C INSIDE	0.000	0.000	
CENTER	0.000	0.000	
FLOWPATH WIDTH-MM	1906.6667		
FLOWPATH LENGTH-MM	2080.0000	(MINIMUM LENGTH)	

TRAY NO.	17 - 19	17 - 19	17 - 19
HEADINGS	#19(110%)	#19(100%)	#19(30%)
TRAY SPACING-MM	600.000	600.000	600.000
UNDERDOWNFLOW-MM	20.000	20.000	20.000
INLET WEIR HEIGHT-MM	0.000	0.000	0.000
EXIT WEIR HEIGHT-MM	25.000	25.000	25.000
PERCENT BLOCK	90.0(1)	90.0(1)	90.0(1)
VALVES PER TRAY	191	191	191
LIGHT VALVES PER TRAY	57	57	57
PERCENT HOLE AREA	4.8	4.8	4.8

NOTES:
(1) GASKET SHOULD BE USED FOR #17-#19 TRAYS.

DESIGN FOR QUOTE/(ORDER) APPROVED BY E. Y. Lee

HANBAL VALVE TRAY DESIGN PROGRAM VERSION 1.0.0T PROCESS DATA 15-Dec-2004 10:55 AM

CLIENT: NATIONAL IRANIAN OIL COMPANY
COLUMN: 103-C-101/CONDENSATE STABILIZER
CASE: SUMMER CASE

DESIGNER: D.S.OHK
REVISED: 0

TRAY NO.	17 - 19	17 - 19	17 - 19
HEADINGS	#19(110%)	#19(100%)	#19(30%)
VAPOR FLOW-KGS/HR	21494.0	19540.0	5862.0
VLOAD-M3/HR	231.44	210.40	63.12
DENSITY-KGS/M3	13.53545	13.53545	13.53545
LIQUID FLOW-KGS/HR	179.3	163.0	48.9
FLOW-M3/HR	0.28	0.25	0.08
DENSITY-KGS/M3	650.772	650.772	650.772
SYSTEM FACTOR-VAPOR	1.000	1.000	1.000
SYSTEM FACTOR-LIQUID	1.000	1.000	1.000
JET FLOOD PERCENT	10.0	9.1	2.7
D/C BACKUP FLOOD PERCENT	20.6	19.2	8.2
D/C LOAD PERCENT SAFE DESIGN	0.2	0.2	0.1
WEEP POINT PERCENT OF DESIGN	12.3	13.5	44.7
P.D./TRAY-KG/CM2	0.002370	0.002211	0.001451
TRAY P.D.-MM.LIQ.	36.412	33.969	22.297
DRY TRAY P.D.-MM.LIQ.	20.423	18.201	8.572
TRAY FOAM HEIGHT-MM.LIQ.	40.486	40.322	39.377
UDFC P.D.-MM.LIQ.	0.001	0.001	0.000
D/C BACKUP-MM.LIQ.	76.899	74.293	61.674
-MM FOAM	76.899	74.293	61.674
FH HOLE VELOCITY FACTOR-M/HR	1014.200	922.000	276.600
FH RECOMMENDED SAFE DESIGN-M/HR	2387.306	2387.306	2387.306
D/C LOADS-M3/M2/HR			
SIDE	1.0	0.9	0.3
OFF-CENTER	0.0	0.0	0.0
CENTER	0.0	0.0	0.0
WEIR LOADS-M3/M/HR			
EFFECTIVE	1.8	1.6	0.5
SUPERFICIAL			
SIDE	0.2	0.2	0.0
O/C OUTSIDE	0.0	0.0	0.0
O/C INSIDE	0.0	0.0	0.0
CENTER	0.0	0.0	0.0
HEIGHT OVER EXIT WEIR-MM.LIQ	4.133	3.878	1.737
SIDE UDF VELOCITY-M/HR	8.831	8.028	2.408

HANBAL VALVE TRAY DESIGN PROGRAM VERSION 1.0.0T PROCESS DATA 15-Dec-2004 10:55 AM

CLIENT: NATIONAL IRANIAN OIL COMPANY
COLUMN: 103-C-101/CONDENSATE STABILIZER
CASE: WINTER CASE

DESIGNER: D.S.OHK
REVISED: 0

TRAY NO.	17 - 19	17 - 19	17 - 19
HEADINGS	#19(110%)	#19(100%)	#19(30%)
VAPOR FLOW-KGS/HR	26109.6	23736.0	7120.8
VLOAD-M3/HR	277.51	252.29	75.69
DENSITY-KGS/M3	13.96821	13.96821	13.96821
LIQUID FLOW-KGS/HR	148.5	135.0	40.5
FLOW-M3/HR	0.23	0.21	0.06
DENSITY-KGS/M3	647.674	647.674	647.674
SYSTEM FACTOR-VAPOR	1.000	1.000	1.000
SYSTEM FACTOR-LIQUID	1.000	1.000	1.000
JET FLOOD PERCENT	12.3	11.2	3.4
D/C BACKUP FLOOD PERCENT	23.6	21.9	9.2
D/C LOAD PERCENT SAFE DESIGN	0.2	0.2	0.1
WEEP POINT PERCENT OF DESIGN	10.2	11.2	37.2
P.D./TRAY-KG/CM2	0.002694	0.002475	0.001457
TRAY P.D.-MM.LIQ.	41.601	38.205	22.497
DRY TRAY P.D.-MM.LIQ.	26.028	22.833	8.989
TRAY FOAM HEIGHT-MM.LIQ.	40.187	40.057	39.340
UDFC P.D.-MM.LIQ.	0.001	0.001	0.000
D/C BACKUP-MM.LIQ.	81.789	78.263	61.837
-MM FOAM	81.789	78.263	61.837
FH HOLE VELOCITY FACTOR-M/HR	1216.127	1105.570	331.671
FH RECOMMENDED SAFE DESIGN-M/HR	2387.306	2387.306	2387.306
D/C LOADS-M3/M2/HR			
SIDE	0.8	0.8	0.2
OFF-CENTER	0.0	0.0	0.0
CENTER	0.0	0.0	0.0
WEIR LOADS-M3/M/HR			
EFFECTIVE	1.5	1.3	0.4
SUPERFICIAL			
SIDE	0.1	0.1	0.0
O/C OUTSIDE	0.0	0.0	0.0
O/C INSIDE	0.0	0.0	0.0
CENTER	0.0	0.0	0.0
HEIGHT OVER EXIT WEIR-MM.LIQ	3.656	3.431	1.537
SIDE UDF VELOCITY-M/HR	7.349	6.681	2.004

GS/ OIEC/ IOEC	STRENGTH CALCULATION SHEETS	N.I.O.C Pars Oil & Gas Company

EQUIPMENT NAME: **COLUMN INTERNALS FOR CONDENSATE STABILISER**

DOCUMENT TITLE: **STRENGTH CALCULATION SHEETS**

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	PRE'D	CHK'D	APP'D
F	Jun.19 '06	FINAL	Y.P.KIM	K.T.MOON	J.H.KWON
1	Jul.23 '05	REVISED BY COMMENTS	Y.P.KIM	K.T.MOON	J.H.KWON
0	June.14'05	FOR APPROVAL	Y.P.KIM	K.T.MOON	J.H.KWON

SOUTH PARS GAS FIELD DEVELOPMENT (PHASES 9&10), ASSALUYEH, IRAN ONSHORE FACILITIES	CONTRACT NO.: POGC-760-81-156
	REQ'N NO.: DI-6340S-0510-LG-0040
HMT	ITEM NO.: 103-C-101/201
	DOCUMENT NO.: VP-6340S-103-CI101-002

VENDOR NAME: HANBAL MASSTECH LIMITED

1. INPUT DATA for 103-C-101/201(CHIMNEY TRAY SUP'T RING)

1)	SHELL INDIA	(SD)= 2600 mm
2)	SUPPORT RING WIDTH	(SW)= 65 mm
3)	SUPPORT USE TH'K	(ST)= 16 mm
4)	SUPPORT RING MAT'L	= A516-60N
5)	CORROSION ALLOWANCE	(CA)= 3 mm
6)	TRAY MAT'L TH'K	(TT)= 3 mm
7)	TRAY MAT'L	= A240-304L
8)	TRAY MAT'L GRAVITY	(TG)= 7.93
9)	TRAY PLATE WEIGHT	(PW)= 176.8313 kg
10)	LIQUID HEAD	(LH)= 1050 mm
11)	LIQUID GRAVITY	(LG)= .75
12)	SPECIFIED UNIT LOAD	(UW)= .0003 Kg/mm²
13)	DESIGN TEMPERATURE	= 225 °C (437 °F)
14)	ALLOWABLE STRESS AT DESIGN TEMP.	(σa)= 17.1 Ksi (12.0212 Kg/mm²)
15)	BEAM Q'TY	(BQ)= 0

2. CALCULATION

1) SHELL SECTIONAL AREA (LA)

$LA = (SD / 1000)^2 \times \pi / 4$

$= 5.309292\ M^2$

2) TRAY PLATE WEIGHT ON LOAD ZONE (W_P)

$W_P = LA \times TT \times TG \times 1.4$

$= 176.8313\ Kg$

3) LIQUID WEIGHT ON LOAD ZONE (W_L)

$W_L = LA \times LH \times LG$

$= 4181.067\ Kg$

4) TOTAL WEIGHT ON LOAD ZONE (W_T)

$W_T = W_P + W_L$

$= 4357.898\ Kg$

5) UNIT LOAD (UW)

$UW = W_T / LA \times 10^{-6}$

$= 4357.898 / 5.309292 \times 10^{-6}$

$= .00082\ Kg/mm^2$

6) REQUIRED THICKNESS

Reference : PRESSURE VESSEL DESIGN MANUAL by DENNIS R. MOSS

$W3 = UW \times SD / 2$ (Max. unit line load)

$= .00082 \times 2600 / 2$

$= 1.0659\ Kg/mm$

$M = W3 \times SW$ (Max. moment per unit length)

$= 1.0659 \times 65$

$= 69.2835\ Kg\text{-}mm/mm$

$RT = \sqrt{(6 \times M / \sigma a)} + 2 \times CA$

$= \sqrt{(6 \times 69.2835 / 12.0212)} + 2 \times 3$

$= 11.8805\ mm$

=====>>> RECOMMENDED SUP'T RING THICKNESS = 16 mm <<<=====

1. INPUT DATA for 103-C-101/201(#2 THRU #15 TRAY SUP'T RING)

1)	SHELL INDIA	(SD)= 2600 mm
2)	SUPPORT RING WIDTH	(SW)= 65 mm
3)	SUPPORT USE TH'K	(ST)= 16 mm
4)	SUPPORT RING MAT'L	= A516-60N
5)	CORROSION ALLOWANCE	(CA)= 3 mm
6)	TRAY MAT'L TH'K	(TT)= 2 mm
7)	TRAY MAT'L	= A240-304L
8)	TRAY MAT'L GRAVITY	(TG)= 7.93
9)	TRAY PLATE WEIGHT	(PW)= 117.8875 kg
10)	LIQUID HEAD	(LH)= 300 mm
11)	LIQUID GRAVITY	(LG)= .75
12)	SPECIFIED UNIT LOAD	(UW)= .0003 Kg/mm²
13)	DESIGN TEMPERATURE	= 225 °C (437 °F)
14)	ALLOWABLE STRESS AT DESIGN TEMP.	(σa)= 17.1 Ksi (12.0212 Kg/mm²)
15)	BEAM Q'TY	(BQ)= 0

2. CALCULATION

1) SHELL SECTIONAL AREA (LA)
   LA = (SD / 1000)² x π / 4
   = 5.309292 M²
2) TRAY PLATE WEIGHT ON LOAD ZONE (W_P)
   W_P = LA x TT x TG x 1.4
   = 117.8875 Kg
3) LIQUID WEIGHT ON LOAD ZONE (W_L)
   W_L = LA x LH x LG
   = 1194.591 Kg
4) TOTAL WEIGHT ON LOAD ZONE (W_T)
   W_T = W_P + W_L
   = 1312.478 Kg
5) UNIT LOAD (UW)
   UW = W_T / LA x 10^{-6}
   = 1312.478 / 5.309292 x 10^{-6}
   = .000247 Kg/mm², But use SPECIFIED UNIT LOAD .0003 Kg/mm²

6) REQUIRED THICKNESS

Reference : PRESSURE VESSEL DESIGN MANUAL by DENNIS R. MOSS

W3 = UW x SD / 2 (Max. unit line load)
= .0003 x 2600 / 2
= .39 Kg/mm
M = W3 x SW (Max. moment per unit length)
= .39 x 65
= 25.3499 Kg-mm/mm
RT = √(6 x M / σa) + 2 x CA
= √(6 x 25.3499 / 12.0212) + 2 x 3
= 9.557 mm

=====>>> RECOMMENDED SUP'T RING THICKNESS = 16 mm <<<=====

1. INPUT DATA for 103-C-101/201(#16,#17,#18,#19 TRAY SUP'T RING)

1)	SHELL INDIA	(SD)= 2600 mm
2)	SUPPORT RING WIDTH	(SW)= 65 mm
3)	SUPPORT USE TH'K	(ST)= 16 mm
4)	SUPPORT RING MAT'L	= A516-60N
5)	CORROSION ALLOWANCE	(CA)= 3 mm
6)	TRAY MAT'L TH'K	(TT)= 2 mm
7)	TRAY MAT'L	= A240-321
8)	TRAY MAT'L GRAVITY	(TG)= 7.93
9)	TRAY PLATE WEIGHT	(PW)= 117.8875 kg
10)	LIQUID HEAD	(LH)= 300 mm
11)	LIQUID GRAVITY	(LG)= .75
12)	SPECIFIED UNIT LOAD	(UW)= .0003 Kg/mm²
13)	DESIGN TEMPERATURE	= 225 °C (437 °F)
14)	ALLOWABLE STRESS AT DESIGN TEMP.	(σa)= 17.1 Ksi (12.0212 Kg/mm²)
15)	BEAM Q'TY	(BQ)= 0

2. CALCULATION

1) SHELL SECTIONAL AREA (LA)

$LA = (SD / 1000)^2 \times \pi / 4$

$= 5.309292\ M^2$

2) TRAY PLATE WEIGHT ON LOAD ZONE (W_P)

$W_P = LA \times TT \times TG \times 1.4$

$= 117.8875\ Kg$

3) LIQUID WEIGHT ON LOAD ZONE (W_L)

$W_L = LA \times LH \times LG$

$= 1194.591\ Kg$

4) TOTAL WEIGHT ON LOAD ZONE (W_T)

$W_T = W_P + W_L$

$= 1312.478\ Kg$

5) UNIT LOAD (UW)

$UW = W_T / LA \times 10^{-6}$

$= 1312.478 / 5.309292 \times 10^{-6}$

$= .000247\ Kg/mm^2$, But use SPECIFIED UNIT LOAD $.0003\ Kg/mm^2$

6) REQUIRED THICKNESS

Reference : PRESSURE VESSEL DESIGN MANUAL by DENNIS R. MOSS

$W3 = UW \times SD / 2$ (Max. unit line load)

$= .0003 \times 2600 / 2$

$= .39\ Kg/mm$

$M = W3 \times SW$ (Max. moment per unit length)

$= .39 \times 65$

$= 25.3499\ Kg\text{-}mm/mm$

$RT = \sqrt{(6 \times M / \sigma a)} + 2 \times CA$

$= \sqrt{(6 \times 25.3499 / 12.0212)} + 2 \times 3$

$= 9.557\ mm$

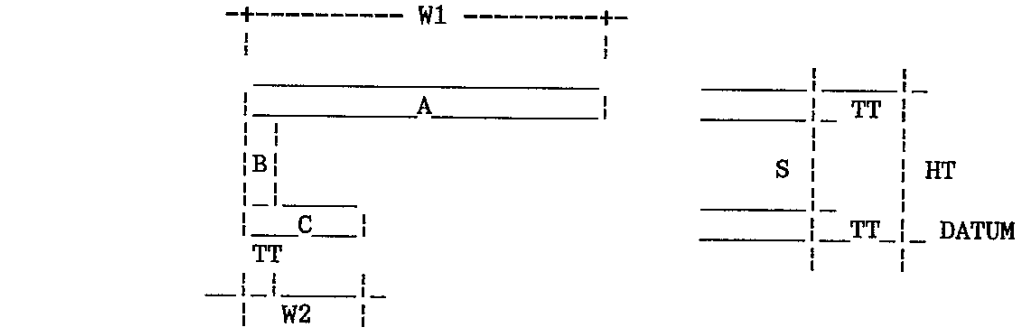
=====>>> RECOMMENDED SUP'T RING THICKNESS = 16 mm <<<=====

1. INPUT DATA FOR 103-C-101/201(ACTIVE DECK EVEN NO. #2 THRU #14)

1) LOADING INFORMATION.

- LOAD WIDTH (WB/WD)= 426 / 0 mm
- LIQUID HEAD (LB/LD)= 130 / 0 mm
- LIQUID GRAVITY (LG)= .75
- PLATE MAT'L = A240-304L (GRAVITY = 7.93)
- MAN LOAD (MW)= 110 Kg
- SPECIFIED UNIT LOAD (UW)= .0002 Kg/mm²
- DESIGN TEMPERATURE = 225 °C (437 °F)
- ALLOWABLE STRESS AT DSN. (σa)= 15.393 Ksi (10.8222 Kg/mm²)
- ALLOWABLE STRESS AT ATM. (σaa)= 16.7 Ksi (11.7412 Kg/mm²)
- MODULUS OF ELAST. AT DSN. (E)= 26.241 x 1000 Ksi (18449.15 Kg/mm²)
- MODULUS OF ELAST. AT ATM. (AE)= 28.3 x 1000 Ksi (19896.77 Kg/mm²)

2) SECTIONAL FIGURE FOR TRAYS



- SPAN LENGTH (L)= 654 mm
- EFFECTIVE DECK WIDTH (W1)= 46.14 mm
- TURNDOWN HEIGHT (HT)= 65 mm
- TURNBACK WIDTH (W2)= 30 mm
- TRAY THICKNESS (TT)= 2 mm
- CORROSION ALLOWANCE (CA)= 0 mm

2. CALCULATION

1) PROPERTIES OF SECTION FOR TRAYS

PART	A [MM2]	Y [MM]	AY [MM3]	d [MM]	d2 [MM2]	Ad2 [MM4]	Ig [MM4]
A	92.3	64.0	5905.9	27.79	772.4	71281.	31.
B	122.0	32.5	3965.0	3.71	13.7	1677.	37830.
C	60.0	1.0	60.0	35.21	1239.5	74373.	20.
TOTAL	274.3	-	9930.9	-	-	147330.	37881.

$I = Ad^2 + Ig = 147330.2 + 37880.93$
$= 185211.2\ mm^4$
$C = AY / A = 9930.92 / 274.28$
$= 36.20723\ mm$
$Z = I / C = 185211.2 / 36.20723$
$= 5115.308\ mm^3$

2) UNIT LOAD

W_L = LB x WB x L x LG x 10^{-6} (Liquid weight)
= 27.1638 Kg

W_P = 6.214647 Kg (Tray weight)

W_T = W_L + W_P = 27.1638 + 6.214647
= 33.37845 Kg

UL = W_T / L = .051 Kg/mm (CALCULATION)
UL = UW x B = .0851 Kg/mm (SPECIFIED)

=====>>> Therefore used SPECIFIED UL value <<<=====

3) STRESS DUE TO MAN LOAD(σm)

σm = MW x L / 4 / Z = 110 x 654 / 4 / 5115.308
= 3.5159 Kg/mm^2 < σaa

4) STRESS DUE TO STATIC LOAD(σs)

σs = UL x L^2 / 8 / Z = .0851 x 654 ^{2} / 8 / 5115.308
= .8894 Kg/mm^2 < σa

=====>>> THEREFORE, O.K ! <<<=====

5) ALLOWABLE DEFLECTION

δa = L / 900 = 654 / 900
= .7267 mm

6) DEFLECTION DUE TO MAN LOAD(δm)

δm = MW x L^3 / 48 / AE / I
= 110 x 654 ^{3} / 48 / 19896.77 / 185211.2
= .1739 mm < δa

7) DEFLECTION DUE TO STATIC LOAD(δs)

δs = 5 x UL x L^4 / 384 / E / I
= 5 x .0851 x 654 ^{4} / 384 / 18449.15 / 185211.2
= .0593 mm < δa

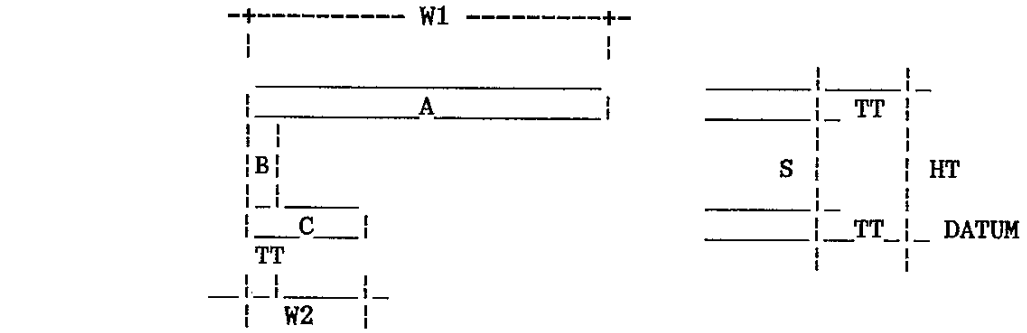
=====>>> THEREFORE, O.K ! <<<=====

1. INPUT DATA FOR 103-C-101/201(ACTIVE DECK ODD NO. #3 THRU #15)

1) LOADING INFORMATION.

- LOAD WIDTH (WB/WD)= 414 / 0 mm
- LIQUID HEAD (LB/LD)= 130 / 0 mm
- LIQUID GRAVITY (LG)= .75
- PLATE MAT'L = A240-304L (GRAVITY = 7.93)
- MAN LOAD (MW)= 110 Kg
- SPECIFIED UNIT LOAD (UW)= .0002 Kg/mm²
- DESIGN TEMPERATURE = 225 °C (437 °F)
- ALLOWABLE STRESS AT DSN. (σa)= 15.393 Ksi (10.8222 Kg/mm²)
- ALLOWABLE STRESS AT ATM. (σaa)= 16.7 Ksi (11.7412 Kg/mm²)
- MODULUS OF ELAST. AT DSN. (E)= 26.241 x 1000 Ksi (18449.15 Kg/mm²)
- MODULUS OF ELAST. AT ATM. (AE)= 28.3 x 1000 Ksi (19896.77 Kg/mm²)

2) SECTIONAL FIGURE FOR TRAYS



- SPAN LENGTH (L)= 604 mm
- EFFECTIVE DECK WIDTH (W1)= 46.14 mm
- TURNDOWN HEIGHT (HT)= 65 mm
- TURNBACK WIDTH (W2)= 30 mm
- TRAY THICKNESS (TT)= 2 mm
- CORROSION ALLOWANCE (CA)= 0 mm

2. CALCULATION

1) PROPERTIES OF SECTION FOR TRAYS

PART	A [MM2]	Y [MM]	AY [MM3]	d [MM]	d2 [MM2]	Ad2 [MM4]	Ig [MM4]
A	92.3	64.0	5905.9	27.79	772.4	71281.	31.
B	122.0	32.5	3965.0	3.71	13.7	1677.	37830.
C	60.0	1.0	60.0	35.21	1239.5	74373.	20.
TOTAL	274.3	-	9930.9	-	-	147330.	37881.

$I = Ad^2 + Ig = 147330.2 + 37880.93$
$= 185211.2\ mm^4$
$C = AY / A = 9930.92 / 274.28$
$= 36.20723\ mm$
$Z = I / C = 185211.2 / 36.20723$
$= 5115.308\ mm^3$

2) UNIT LOAD

W_L = LB x WB x L x LG x 10^{-6} (Liquid weight)
= 24.3804 Kg

W_P = 5.607325 Kg (Tray weight)

W_T = W_L + W_P = 24.3804 + 5.607325
= 29.98772 Kg

UL = W_T / L = .0496 Kg/mm (CALCULATION)
UL = UW x B = .0828 Kg/mm (SPECIFIED)

=====>>> Therefore used SPECIFIED UL value <<<=====

3) STRESS DUE TO MAN LOAD(σm)

σm = MW x L / 4 / Z = 110 x 604 / 4 / 5115.308
= 3.2471 Kg/mm^2 < σaa

4) STRESS DUE TO STATIC LOAD(σs)

σs = UL x L^2 / 8 / Z = .0828 x 604^2 / 8 / 5115.308
= .7381 Kg/mm^2 < σa

=====>>> THEREFORE, O.K ! <<<=====

5) ALLOWABLE DEFLECTION

δa = L / 900 = 604 / 900
= .6711 mm

6) DEFLECTION DUE TO MAN LOAD(δm)

δm = MW x L^3 / 48 / AE / I
= 110 x 604^3 / 48 / 19896.77 / 185211.2
= .137 mm < δa

7) DEFLECTION DUE TO STATIC LOAD(δs)

δs = 5 x UL x L^4 / 384 / E / I
= 5 x .0828 x 604^4 / 384 / 18449.15 / 185211.2
= .0419 mm < δa

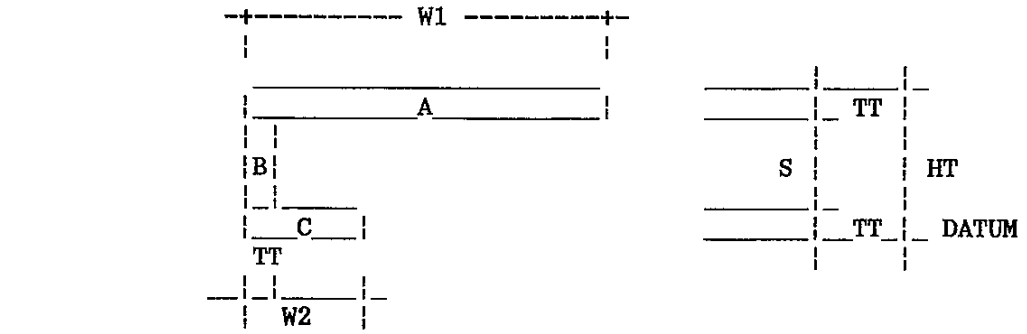
=====>>> THEREFORE, O.K ! <<<=====

1. INPUT DATA FOR 103-C-101/201(ACTIVE DECK #16 TRAY)

1) LOADING INFORMATION.

- LOAD WIDTH (WB/WD)= 426 / 0 mm
- LIQUID HEAD (LB/LD)= 130 / 0 mm
- LIQUID GRAVITY (LG)= .75
- PLATE MAT'L = A240-321 (GRAVITY = 7.93)
- MAN LOAD (MW)= 110 Kg
- SPECIFIED UNIT LOAD (UW)= .0002 Kg/mm²
- DESIGN TEMPERATURE = 225 °C (437 °F)
- ALLOWABLE STRESS AT DSN. (σa)= 18.7 Ksi (13.1473 Kg/mm²)
- ALLOWABLE STRESS AT ATM. (σaa)= 20 Ksi (14.0613 Kg/mm²)
- MODULUS OF ELAST. AT DSN. (E)= 26.241 x 1000 Ksi (18449.15 Kg/mm²)
- MODULUS OF ELAST. AT ATM. (AE)= 28.3 x 1000 Ksi (19896.77 Kg/mm²)

2) SECTIONAL FIGURE FOR TRAYS



- SPAN LENGTH (L)= 654 mm
- EFFECTIVE DECK WIDTH (W1)= 41.862 mm
- TURNDOWN HEIGHT (HT)= 65 mm
- TURNBACK WIDTH (W2)= 30 mm
- TRAY THICKNESS (TT)= 2 mm
- CORROSION ALLOWANCE (CA)= 0 mm

2. CALCULATION

1) PROPERTIES OF SECTION FOR TRAYS

PART	A [MM2]	Y [MM]	AY [MM3]	d [MM]	d2 [MM2]	Ad2 [MM4]	Ig [MM4]
A	83.7	64.0	5358.3	28.69	823.0	68903.	28.
B	122.0	32.5	3965.0	2.81	7.9	965.	37830.
C	60.0	1.0	60.0	34.31	1177.3	70640.	20.
TOTAL	265.7	-	9383.3	-	-	140508.	37878.

$I = Ad^2 + Ig = 140508.5 + 37878.07$
$= 178386.5\ mm^4$
$C = AY / A = 9383.336 / 265.724$
$= 35.31234\ mm$
$Z = I / C = 178386.5 / 35.31234$
$= 5051.677\ mm^3$

2) UNIT LOAD

$W_L = LB \times WB \times L \times LG \times 10^{-6}$ (Liquid weight)
$= 27.1638$ Kg

$W_P = 6.214647$ Kg (Tray weight)

$W_T = W_L + W_P = 27.1638 + 6.214647$
$= 33.37845$ Kg

$UL = W_T / L = .051$ Kg/mm (CALCULATION)
$UL = UW \times B = .0851$ Kg/mm (SPECIFIED)

=====>>> Therefore used SPECIFIED UL value <<<=====

3) STRESS DUE TO MAN LOAD(σm)

$\sigma m = MW \times L / 4 / Z = 110 \times 654 / 4 / 5051.677$
$= 3.5602$ Kg/mm^2 < σaa

4) STRESS DUE TO STATIC LOAD(σs)

$\sigma s = UL \times L^2 / 8 / Z = .0851 \times 654^2 / 8 / 5051.677$
$= .9006$ Kg/mm^2 < σa

=====>>> THEREFORE, O.K ! <<<=====

5) ALLOWABLE DEFLECTION

$\delta a = L / 900 = 654 / 900$
$= .7267$ mm

6) DEFLECTION DUE TO MAN LOAD(δm)

$\delta m = MW \times L^3 / 48 / AE / I$
$= 110 \times 654^3 / 48 / 19896.77 / 178386.5$
$= .1806$ mm < δa

7) DEFLECTION DUE TO STATIC LOAD(δs)

$\delta s = 5 \times UL \times L^4 / 384 / E / I$
$= 5 \times .0851 \times 654^4 / 384 / 18449.15 / 178386.5$
$= .0615$ mm < δa

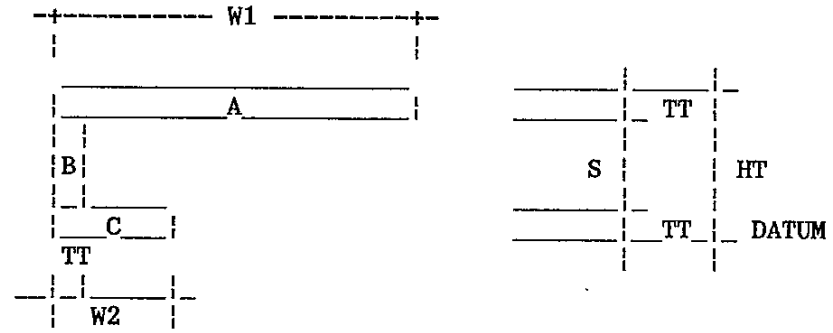
=====>>> THEREFORE, O.K ! <<<=====

1. INPUT DATA FOR 103-C-101/201(ACTIVE DECK #17, #18, #19)

1) LOADING INFORMATION.

- LOAD WIDTH (WB/WD)= 397 / 0 mm
- LIQUID HEAD (LB/LD)= 75 / 0 mm
- LIQUID GRAVITY (LG)= .75
- PLATE MAT'L = A240-321 (GRAVITY = 7.93)
- MAN LOAD (MW)= 110 Kg
- SPECIFIED UNIT LOAD (UW)= .0002 Kg/mm²
- DESIGN TEMPERATURE = 225 °C (437 °F)
- ALLOWABLE STRESS AT DSN. (σa)= 18.7 Ksi (13.1473 Kg/mm²)
- ALLOWABLE STRESS AT ATM. (σaa)= 20 Ksi (14.0613 Kg/mm²)
- MODULUS OF ELAST. AT DSN. (E)= 26.241 x 1000 Ksi (18449.15 Kg/mm²)
- MODULUS OF ELAST. AT ATM. (AE)= 28.3 x 1000 Ksi (19896.77 Kg/mm²)

2) SECTIONAL FIGURE FOR TRAYS



- SPAN LENGTH (L)= 2004 mm
- EFFECTIVE DECK WIDTH (W1)= 41.862 mm
- TURNDOWN HEIGHT (HT)= 100 mm
- TURNBACK WIDTH (W2)= 50 mm
- TRAY THICKNESS (TT)= 2 mm
- CORROSION ALLOWANCE (CA)= 0 mm

2. CALCULATION

1) PROPERTIES OF SECTION FOR TRAYS

PART	A [MM2]	Y [MM]	AY [MM3]	d [MM]	d2 [MM2]	Ad2 [MM4]	Ig [MM4]
A	83.7	99.0	8288.7	51.12	2613.5	218815.	28.
B	192.0	50.0	9600.0	2.12	4.5	865.	147456.
C	100.0	1.0	100.0	46.88	2197.5	219749.	33.
TOTAL	375.7	-	17988.7	-	-	439428.	147517.

$I = Ad^2 + Ig = 439428.5 + 147517.2$
$= 586945.7\ mm^4$
$C = AY / A = 17988.68 / 375.724$
$= 47.87737\ mm$
$Z = I / C = 586945.7 / 47.87737$
$= 12259.36\ mm^3$

2) UNIT LOAD

W_L = LB x WB x L x LG x 10^{-6} (Liquid weight)
= 44.7518 Kg

W_P = 19.99337 Kg (Tray weight)

W_T = W_L + W_P = 44.7518 + 19.99337
= 64.74517 Kg

UL = W_T / L = .0323 Kg/mm (CALCULATION)
UL = UW x B = .0793 Kg/mm (SPECIFIED)

=====>>> Therefore used SPECIFIED UL value <<<=====

3) STRESS DUE TO MAN LOAD(σm)

σm = MW x L / 4 / Z = 110 x 2004 / 4 / 12259.36
= 4.4953 Kg/mm^2 < σaa

4) STRESS DUE TO STATIC LOAD(σs)

σs = UL x L^2 / 8 / Z = .0793 x 2004^2 / 8 / 12259.36
= 3.2472 Kg/mm^2 < σa

=====>>> THEREFORE, O.K ! <<<=====

5) ALLOWABLE DEFLECTION

δa = L / 900 = 2004 / 900
= 2.2267 mm

6) DEFLECTION DUE TO MAN LOAD(δm)

δm = MW x L^3 / 48 / AE / I
= 110 x 2004^3 / 48 / 19896.77 / 586945.7
= 1.5792 mm < δa

7) DEFLECTION DUE TO STATIC LOAD(δs)

δs = 5 x UL x L^4 / 384 / E / I
= 5 x .0793 x 2004^4 / 384 / 18449.15 / 586945.7
= 1.5379 mm < δa

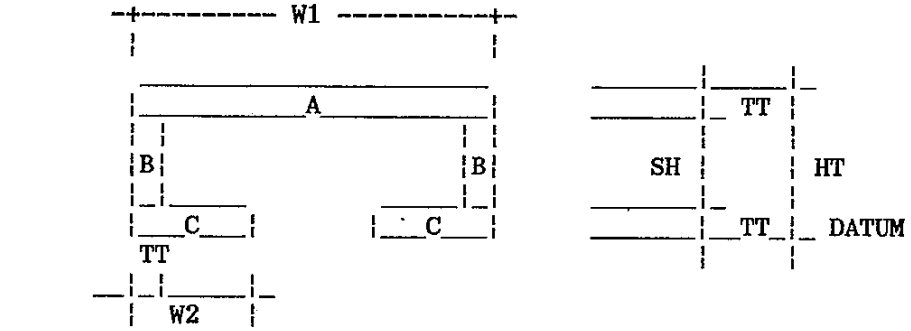
=====>>> THEREFORE, O.K ! <<<=====

1. INPUT DATA FOR 103-C-101/201(CENTER DECK #2 THRU #14)

1) LOADING INFORMATION.

- LOAD WIDTH (WB/WD)= 654 / 316 mm
- LIQUID HEAD (LB/LD)= 130 / 450 mm
- LIQUID GRAVITY (LG)= .75
- PLATE MAT'L = A240-304L (GRAVITY = 7.93)
- MAN LOAD (MW)= 110 Kg
- SPECIFIED UNIT LOAD (UW)= .0003 Kg/mm²
- DESIGN TEMPERATURE = 225 °C (437 °F)
- ALLOWABLE STRESS AT DSN. (σa)= 15.393 Ksi (10.8222 Kg/mm²)
- ALLOWABLE STRESS AT ATM. (σaa)= 16.7 Ksi (11.7412 Kg/mm²)
- MODULUS OF ELAST. AT DSN. (E)= 26.241 x 1000 Ksi (18449.15 Kg/mm²)
- MODULUS OF ELAST. AT ATM. (AE)= 28.3 x 1000 Ksi (19896.77 Kg/mm²)

2) SECTIONAL FIGURE FOR TRAYS



- SPAN LENGTH (L)= 2581 mm
- EFFECTIVE DECK WIDTH (W1)= 380 mm
- TURNDOWN HEIGHT (HT)= 120 mm
- TURNBACK WIDTH (W2)= 65 mm
- TRAY THICKNESS (TT)= 2 mm
- CORROSION ALLOWANCE (CA)= 0 mm

2. CALCULATION

1) PROPERTIES OF SECTION FOR TRAYS

PART	A [MM2]	Y [MM]	AY [MM3]	d [MM]	d2 [MM2]	Ad2 [MM4]	Ig [MM4]
A	760.0	119.0	90440.0	38.91	1513.6	1150359.	253.
B	464.0	60.0	27840.0	20.09	403.8	187360.	520299.
C	260.0	1.0	260.0	79.09	6256.0	1626548.	87.
TOTAL	1480.0	-	118540.0	-	-	2938607.	520637.

$I = Ad^2 + Ig = 2938607 + 520637.3$
$= 3459244\ mm^4$
$C = AY / A = 118540 / 1480$
$= 80.0946\ mm$
$Z = I / C = 3459244 / 80.0946$
$= 43189.48\ mm^3$

2) UNIT LOAD

W_L = (LBxWB + LDxWD) x L x LG x 10^{-6} (Liquid weight)
= 439.841 Kg

W_P = 63.08031 Kg (Tray weight)

W_T = W_L + W_P = 439.841 + 63.08031
= 502.9213 Kg

UL = W_T / L = .1948 Kg/mm (CALCULATION)
UL = UW x B = .291 Kg/mm (SPECIFIED)

=====>>> Therefore used SPECIFIED UL value <<<=====

3) STRESS DUE TO MAN LOAD(σm)

σm = MW x L / 4 / Z = 110 x 2581 / 4 / 43189.48
= 1.6433 Kg/mm^2 < σaa

4) STRESS DUE TO STATIC LOAD(σs)

σs = UL x L^2 / 8 / Z = .291 x 2581 ^{2} / 8 / 43189.48
= 5.6104 Kg/mm^2 < σa

=====>>> THEREFORE, O.K ! <<<=====

5) ALLOWABLE DEFLECTION

δa = L / 900 = 2581 / 900
= 2.8678 mm

6) DEFLECTION DUE TO MAN LOAD(δm)

δm = MW x L^3 / 48 / AE / I
= 110 x 2581 ^{3} / 48 / 19896.77 / 3459244
= .5724 mm < δa

7) DEFLECTION DUE TO STATIC LOAD(δs)

δs = 5 x UL x L^4 / 384 / E / I
= 5 x .291 x 2581 ^{4} / 384 / 18449.15 / 3459244
= 2.6346 mm < δa

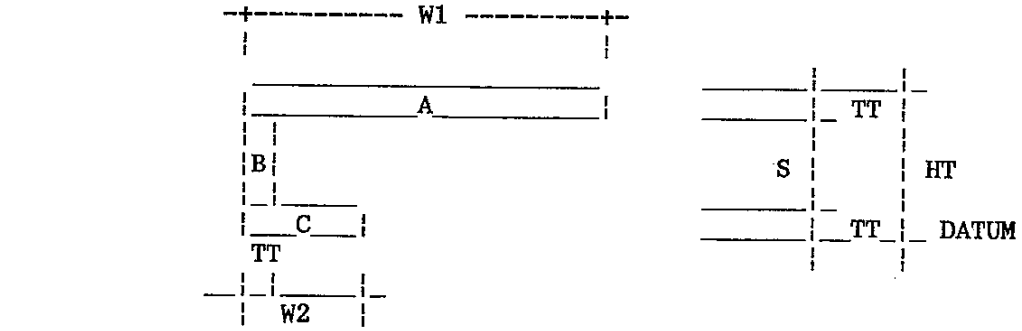
=====>>> THEREFORE, O.K ! <<<=====

1. INPUT DATA FOR 103-C-101/201(#1 CHIMNEY TRAY PLATE)

1) LOADING INFORMATION.

- LOAD WIDTH	(WB/WD)=	0 / 951 mm
- LIQUID HEAD	(LB/LD)=	0 / 1050 mm
- LIQUID GRAVITY	(LG)=	.75
- PLATE MAT'L	=	A240-304L (GRAVITY = 7.93)
- MAN LOAD	(MW)=	110 Kg
- SPECIFIED UNIT LOAD	(UW)=	.0003 Kg/mm²
- DESIGN TEMPERATURE	=	225 °C (437 °F)
- ALLOWABLE STRESS AT DSN.	(σa)=	15.393 Ksi (10.8222 Kg/mm²)
- ALLOWABLE STRESS AT ATM.	(σaa)=	16.7 Ksi (11.7412 Kg/mm²)
- MODULUS OF ELAST. AT DSN.	(E)=	26.241 x 1000 Ksi (18449.15 Kg/mm²)
- MODULUS OF ELAST. AT ATM.	(AE)=	28.3 x 1000 Ksi (19896.77 Kg/mm²)

2) SECTIONAL FIGURE FOR TRAYS



- SPAN LENGTH	(L)=	2021 mm
- EFFECTIVE DECK WIDTH	(W1)=	69.21 mm
- TURNDOWN HEIGHT	(HT)=	160 mm
- TURNBACK WIDTH	(W2)=	50 mm
- TRAY THICKNESS	(TT)=	3 mm
- CORROSION ALLOWANCE	(CA)=	0 mm

2. CALCULATION

1) PROPERTIES OF SECTION FOR TRAYS

PART	A [MM2]	Y [MM]	AY [MM3]	d [MM]	d2 [MM2]	Ad2 [MM4]	Ig [MM4]
A	207.6	158.5	32909.4	72.98	5326.2	1105869.	156.
B	462.0	80.0	36960.0	5.52	30.5	14075.	913066.
C	150.0	1.5	225.0	84.02	7059.3	1058892.	113.
TOTAL	819.6	-	70094.4	-	-	2178836.	913334.

$I = Ad^2 + Ig = 2178836 + 913334.3$
$= 3092170\ mm^4$
$C = AY / A = 70094.36 / 819.63$
$= 85.51952\ mm$
$Z = I / C = 3092170 / 85.51952$
$= 36157.48\ mm^3$

2) UNIT LOAD

W_L = (LBxWB + LDxWD) / 2 x L x LG x 10^{-6} (Liquid weight)
= 756.7761 Kg

W_P = 37.90234 Kg (Tray weight)

W_T = W_L + W_P = 756.7761 + 37.90234
= 794.6784 Kg

UL = W_T / L = .3932 Kg/mm (CALCULATION)
UL = UW x B = .2853 Kg/mm (SPECIFIED)

=====>>> Therefore used CALCULATOIN UL value <<<=====

3) STRESS DUE TO MAN LOAD(σm)

σm = MW x L / 4 / Z = 110 x 2021 / 4 / 36157.48
= 1.537 Kg/mm^2 < σaa

4) STRESS DUE TO STATIC LOAD(σs)

σs = UL x L^2 / 8 / Z = .3932 x 2021 ^{2} / 8 / 36157.48
= 5.5521 Kg/mm^2 < σa

=====>>> THEREFORE, O.K ! <<<=====

5) ALLOWABLE DEFLECTION

δa = L / 900 = 2021 / 900
= 2.2455 mm

6) DEFLECTION DUE TO MAN LOAD(δm)

δm = MW x L^3 / 48 / AE / I
= 110 x 2021 ^{3} / 48 / 19896.77 / 3092170
= .3074 mm < δa

7) DEFLECTION DUE TO STATIC LOAD(δs)

δs = 5 x UL x L^4 / 384 / E / I
= 5 x .3932 x 2021 ^{4} / 384 / 18449.15 / 3092170
= 1.4971 mm < δa

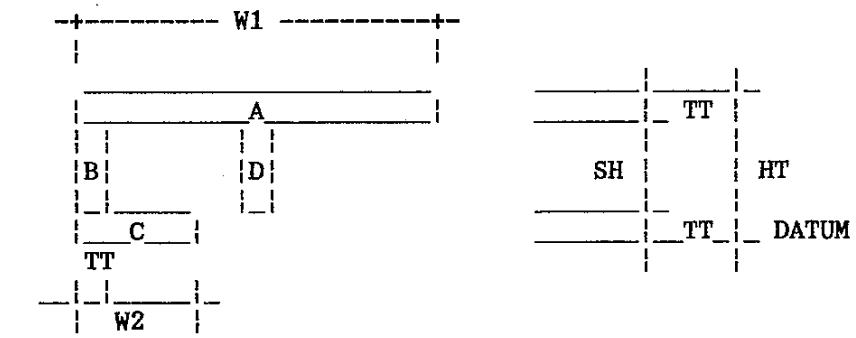
=====>>> THEREFORE, O.K ! <<<=====

1. INPUT DATA FOR 103-C-101/201(SIDE DECK #3 THRU #15)

1) LOADING INFORMATION.

- LOAD WIDTH (WB/WD)= 604 / 423 mm
- LIQUID HEAD (LB/LD)= 130 / 450 mm
- LIQUID GRAVITY (LG)= .75
- PLATE MAT'L = A240-304L (GRAVITY = 7.93)
- MAN LOAD (MW)= 110 Kg
- SPECIFIED UNIT LOAD (UW)= .0003 Kg/mm²
- DESIGN TEMPERATURE = 225 °C (437 °F)
- ALLOWABLE STRESS AT DSN. (σa)= 15.393 Ksi (10.8222 Kg/mm²)
- ALLOWABLE STRESS AT ATM. (σaa)= 16.7 Ksi (11.7412 Kg/mm²)
- MODULUS OF ELAST. AT DSN. (E)= 26.241 x 1000 Ksi (18449.15 Kg/mm²)
- MODULUS OF ELAST. AT ATM. (AE)= 28.3 x 1000 Ksi (19896.77 Kg/mm²)

2) SECTIONAL FIGURE FOR TRAYS



- SPAN LENGTH (L)= 1920 mm
- EFFECTIVE DECK WIDTH (W1)= 180 mm
- TURNDOWN HEIGHT (HT)= 120 mm (HD)= 100 mm
- TURNBACK WIDTH (W2)= 50 mm
- TRAY THICKNESS (TT)= 2 mm
- CORROSION ALLOWANCE (CA)= 0 mm

2. CALCULATION

1) PROPERTIES OF SECTION FOR TRAYS

PART	A [MM2]	Y [MM]	AY [MM3]	d [MM]	d2 [MM2]	Ad2 [MM4]	Ig [MM4]
A	360.0	119.0	42840.0	39.30	1544.3	555940.	120.
B	232.0	60.0	13920.0	19.70	388.2	90062.	260149.
C	100.0	1.0	100.0	78.70	6194.1	619412.	33.
D	196.0	71.0	13916.0	8.70	75.7	14844.	156865.
TOTAL	888.0	-	70776.0	-	-	1280258.	417168.

$I = Ad^2 + Ig = 1280258 + 417168$
$= 1697426\ mm^4$
$C = AY / A = 70776 / 888$
$= 79.70271\ mm$
$Z = I / C = 1697426 / 79.70271$
$= 21296.96\ mm^3$

2) UNIT LOAD

W_L = (LBxWB + LDxWD) / 2 x L x LG x 10^{-6} (Liquid weight)
= 193.5864 Kg

W_P = 27.43729 Kg (Tray weight)

W_T = W_L + W_P = 193.5864 + 27.43729
= 221.0237 Kg

UL = W_T / L = .1151 Kg/mm (CALCULATION)
UL = UW x B = .3081 Kg/mm (SPECIFIED)

=====>>> Therefore used SPECIFIED UL value <<<=====

3) STRESS DUE TO MAN LOAD(σm)

σm = MW x L / 4 / Z = 110 x 1920 / 4 / 21296.96
= 2.4792 Kg/mm^2 < σaa

4) STRESS DUE TO STATIC LOAD(σs)

σs = UL x L^2 / 8 / Z = .3081 x 1920^2 / 8 / 21296.96
= 6.6663 Kg/mm^2 < σa

=====>>> THEREFORE, O.K ! <<<=====

5) ALLOWABLE DEFLECTION

δa = L / 900 = 1920 / 900
= 2.1333 mm

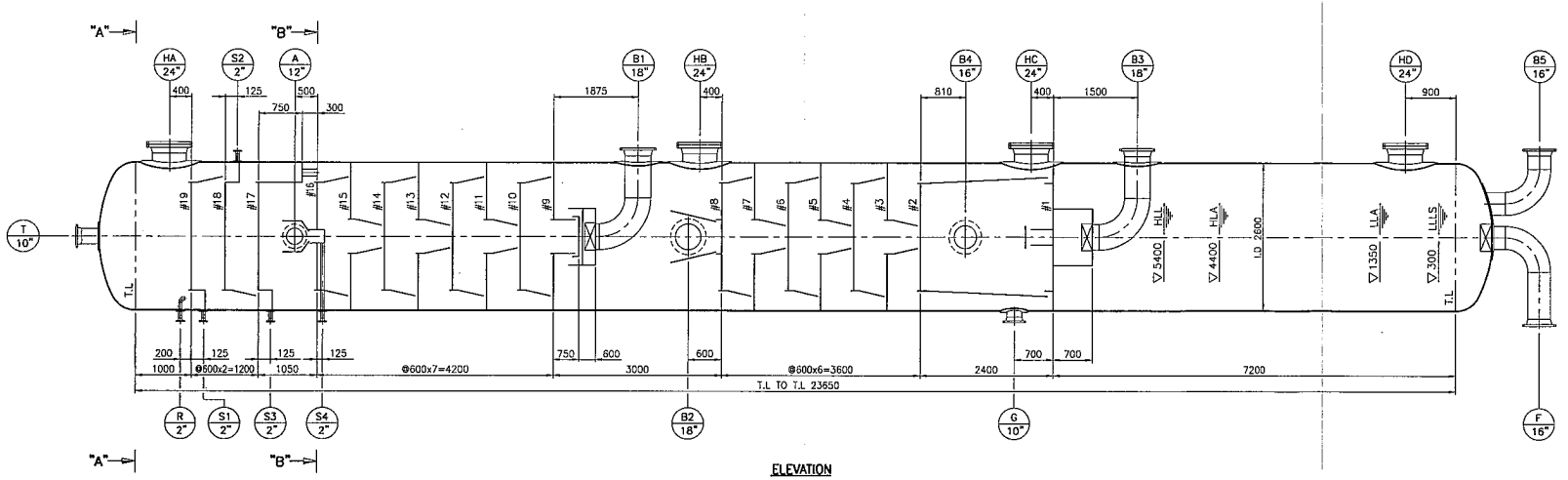
6) DEFLECTION DUE TO MAN LOAD(δm)

δm = MW x L^3 / 48 / AE / I
= 110 x 1920^3 / 48 / 19896.77 / 1697426
= .4802 mm < δa

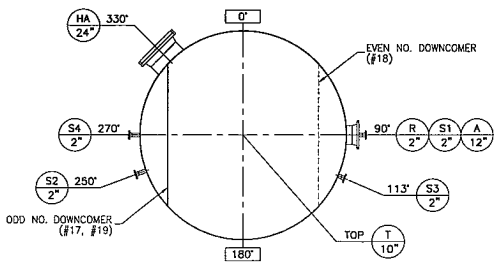
7) DEFLECTION DUE TO STATIC LOAD(δs)

δs = 5 x UL x L^4 / 384 / E / I
= 5 x .3081 x 1920^4 / 384 / 18449.15 / 1697426
= 1.7408 mm < δa

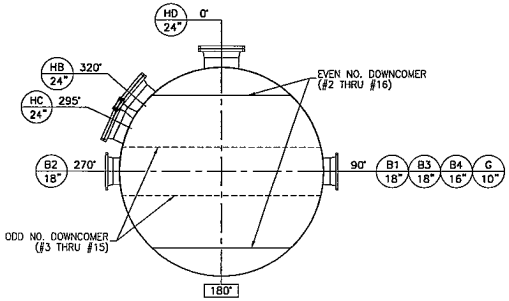
=====>>> THEREFORE, O.K ! <<<=====



ELEVATION



ORIENTATION PLAN
SECTION "A-A"
(#17 THRU #19)



ORIENTATION PLAN
SECTION "B-B"
(#2 THRU #16)

GENERAL INFORMATION

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2. ALL TOWER ATTACHMENTS SHOWN ON THIS DRAWING BE NOT ACTUAL FABRICATION DRAWING. THEREFORE, THE FABRICATION DRAWING SHOULD BE PREPARED BY VESSEL VENDOR.
3. ALL PARTS SHOWN ON THIS DRAWING SHALL BE SUPPLIED AND INSTALLED BY VESSEL FABRICATOR.
4. ALL TRAY ELEVATIONS ARE TO TOP OF THE TRAY SUPPORT RINGS.
5. SUPPORT RINGS ARE TO BE LEVEL ALL AROUND CIRCUMFERENCE OF TOWER, ACROSS TOWER, ACROSS SUPPORT RING FROM TOWER TO INNER EDGE.
6. GRIND OFF ALL BURRS AND SLUG ON THE TOP SURFACE OF SUPPORT RING.
7. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
3	21/02/06	FOR FINAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
2	28/12/05	FOR FINAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
1	27/04/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
0	05/03/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

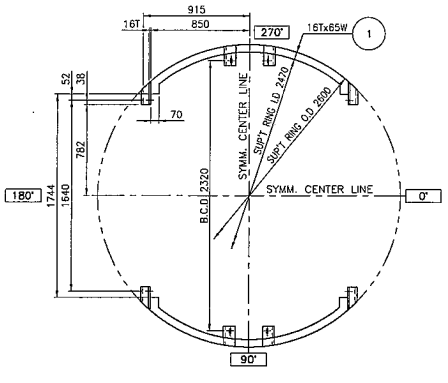
PROJECT No.: POGC-760-81-156 | DOC. CLASS. | SCALE : NONE

HMT | OIEC | IOEC

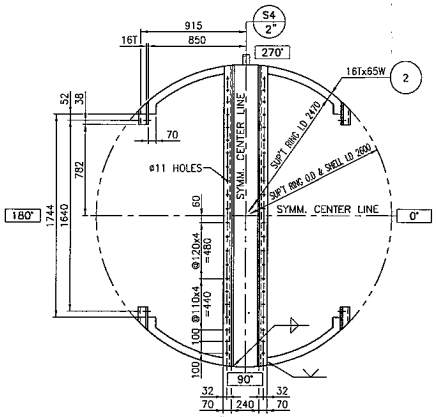
TITLE:
103-C-101/201
CONDENSATE STABILISER
VESSEL SKETCH

DWG No. VP-6340S-103-C1101-003 | SHEET No. 1 OF 11 | REV. F

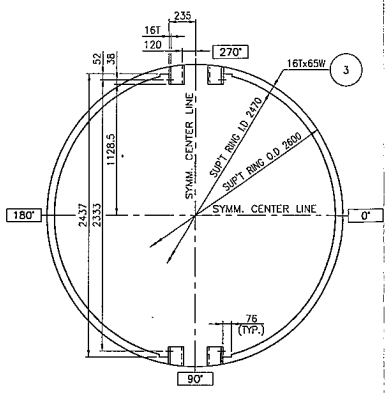
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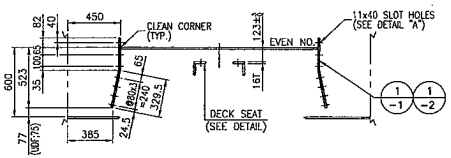
PLAN VIEW OF EVEN NO. TRAY SUPPORT RING(#2 THRU #14)



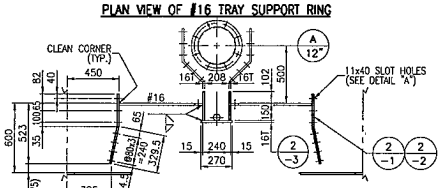
PLAN VIEW OF #16 TRAY SUPPORT RING



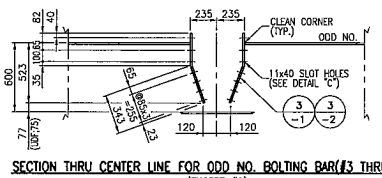
PLAN VIEW OF ODD NO. TRAY SUPPORT RING(#3 THRU #15)
(EXCEPT #9)



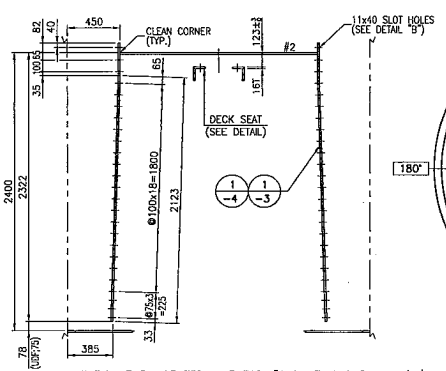
SECTION THRU CENTER LINE FOR EVEN NO. BOLTING BAR(#4 THRU #14)



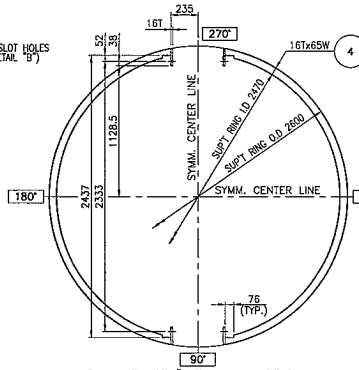
SECTION THRU CENTER LINE FOR #16 BOLTING BAR



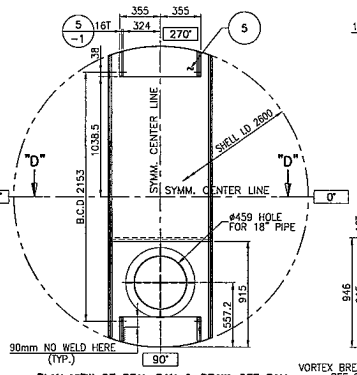
SECTION THRU CENTER LINE FOR ODD NO. BOLTING BAR(#3 THRU #15)
(EXCEPT #9)



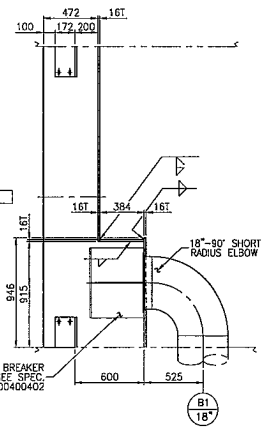
SECTION THRU CENTER LINE FOR #2 BOLTING BAR



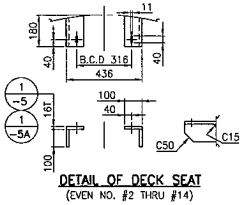
PLAN VIEW OF #9 TRAY SUPPORT RING



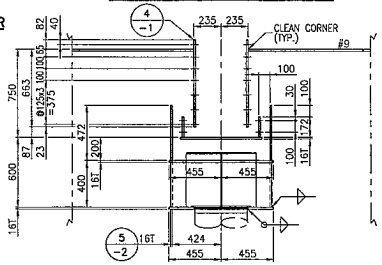
PLAN VIEW OF SEAL PAN & DRAW-OFF PAN



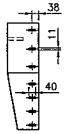
SECTION THRU CENTER LINE FOR SEAL PAN & DRAW-OFF PAN



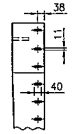
DETAIL OF DECK SEAT
(EVEN NO. #2 THRU #14)



SECTION THRU CENTER LINE FOR #9 BOLTING BAR, SEAL PAN & DRAW-OFF PAN VIEW "D-D"



DETAIL "A"



DETAIL "B"



DETAIL "C"

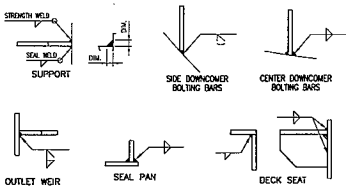
NO.		DESCRIPTION	MAT'L	Q'TY NET	Q'TY UNIT	REMARKS
1		SUPPORT RING	A516GR60N	7	SETS	16Tx65W
	-1	BOLTING BAR	A516GR60N	12	EA	16T
	-2	BOLTING BAR	A516GR60N	12	EA	16T
	-3	BOLTING BAR	A516GR60N	2	EA	16T
	-4	BOLTING BAR	A516GR60N	2	EA	16T
	-5	DECK SEAT	A516GR60N	14	EA	16T
	-5A	DECK SEAT	A516GR60N	14	EA	16T
2		SUPPORT RING	A516GR60N	1	SET	16Tx65W
	-1	BOLTING BAR	A516GR60N	2	EA	16T
	-2	BOLTING BAR	A516GR60N	2	EA	16T
	-3	SUMP	A516GR60N	1	EA	16T
3		SUPPORT RING	A516GR60N	6	SETS	16Tx65W
	-1	BOLTING BAR	A516GR60N	12	EA	16T
	-2	BOLTING BAR	A516GR60N	12	EA	16T
4		SUPPORT RING	A516GR60N	1	SET	16Tx65W
	-1	BOLTING BAR	A516GR60N	4	EA	16T
5		SEAL PAN SUPPORT	A516GR60N	2	SETS	16T
	-1	BOLTING BAR	A516GR60N	4	EA	16T
	-2	DRAW-OFF PAN	A516GR60N	1	SET	16T

(TOTAL; 2 SETS)

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4. ALL TRAY ELEVATIONS ARE TO TOP OF THE TRAY SUPPORT RINGS.
5. SUPPORT RINGS ARE TO BE LEVEL ALL AROUND CIRCUMFERENCE OF TOWER, ACROSS TOWER, ACROSS SUPPORT RING FROM TOWER TO INNER EDGE.
6. GRIND OFF ALL BURRS AND SLUG ON THE TOP SURFACE OF SUPPROT RING.
7. REFERENCE DWG. NO. : VP-6340S-103-C101-003 ~ 013.

TYPICAL WELD DETAILS



NOTE : All welds are continuous 11mm fillet welds unless otherwise specified.

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
I	23/06/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
0	05/03/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

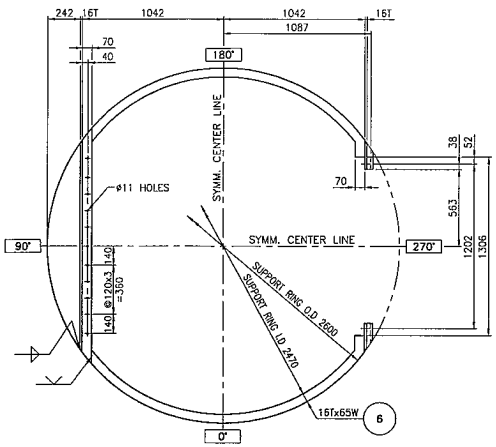
PROJECT No.: POGC-760-81-156	DOC. CLASS.	SCALE : NONE

HMT | OIEC | IOEC

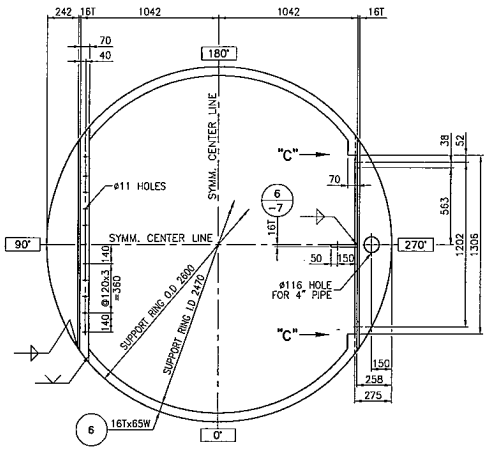
TITLE:
103-C-101/201
CONDENSATE STABILISER
TOWER ATTACHMENT DWG.(1/3)

DWG No.	VP-6340S-103-C101-004	SHEET No. 2 OF 11	REV. F

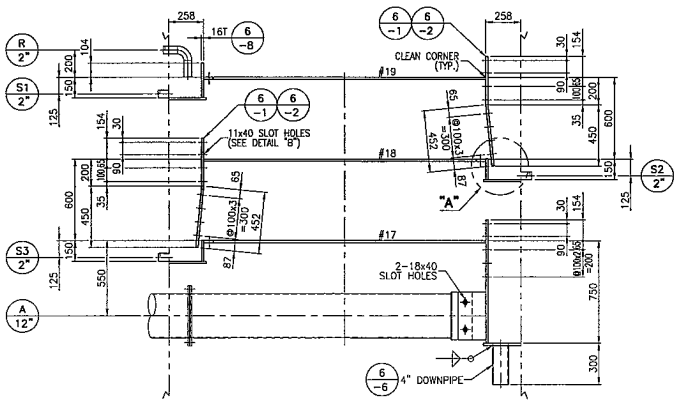
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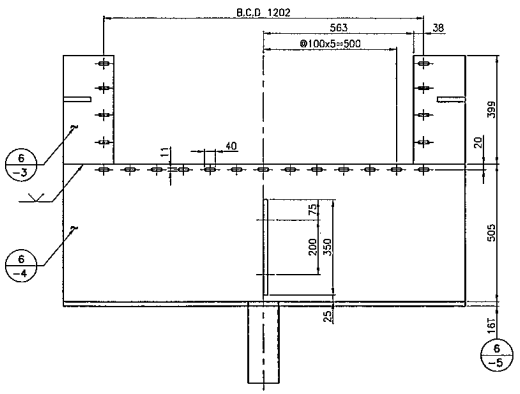
PLAN VIEW OF #19 TRAY SUPPORT RING
ROTATED 180° FOR #18 TRAY SUPPORT RING



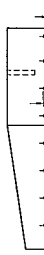
PLAN VIEW OF #17 TRAY SUPPORT RING



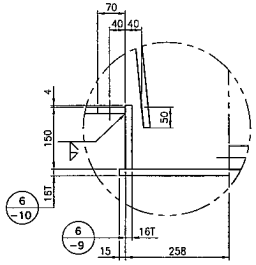
SECTION THRU CENTER LINE FOR #17, #18, #19 BOLTING BAR
& DOWNPIPE



VIEW "C-C"



DETAIL "B"



VIEW "A"

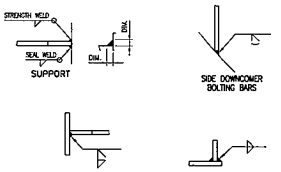
NO.		DESCRIPTION	MAT'L	Q'TY		REMARKS
				MCT	UNIT	
6		SUPPORT RING	A516GR60N	3	SETS	16Tx65W
	-1	BOLTING BAR	A516GR60N	2	EA	16T
	-2	BOLTING BAR	A516GR60N	2	EA	16T
	-3	BOLTING BAR	A516GR60N	2	EA	16T
	-4	BOLTING BAR	A516GR60N	1	EA	16T
	-5	DECK	A516GR60N	1	EA	16T
	-6	DOWNPIPE	A312-TP304L	1	EA	4"
	-7	CLIP	A516GR60N	1	EA	16T
	-8	SIDE SUMP	A516GR60N	1	EA	16T
	-9	SIDE SUMP	A516GR60N	2	EA	16T
	-10	SIDE SUMP	A516GR60N	3	EA	16T

(TOTAL; 2 SETS)

GENERAL INFORMATION

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3. ALL PARTS SHOWN ON THIS DRAWING SHALL BE SUPPLIED AND INSTALLED BY VESSEL FABRICATOR.
4. ALL TRAY ELEVATIONS ARE TO TOP OF THE TRAY SUPPORT RINGS.
5. SUPPORT RINGS ARE TO BE LEVEL ALL AROUND CIRCUMFERENCE OF TOWER, ACROSS TOWER, ACROSS SUPPORT RING FROM TOWER TO INNER EDGE.
6. GRIND OFF ALL BURRS AND SLUG ON THE TOP SURFACE OF SUPPROT RING.
7. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013.

TYPICAL WELD DETAILS



NOTE : All welds are continuous 11mm fillet welds unless otherwise specified.

FINAL
2006. 6. 19
HANBAL MASSTECH
LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.I.Moon	J.H.Kwon	
2	23/06/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.I.Moon	J.H.Kwon	
1	27/04/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.I.Moon	J.H.Kwon	
0	05/03/05	FOR APPROVAL	HMT	Y.P.KIM	K.I.Moon	J.H.Kwon	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

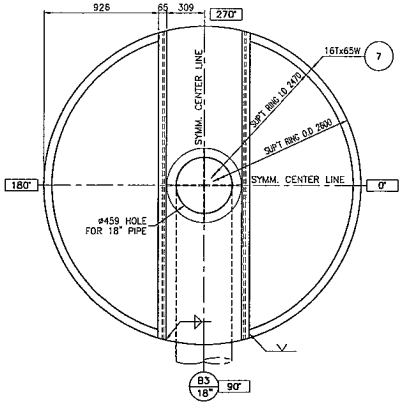
PROJECT No.: POGC-760-81-156 | DOC. CLASS. | SCALE : NONE

HMT | OIEC | I.O.E.C

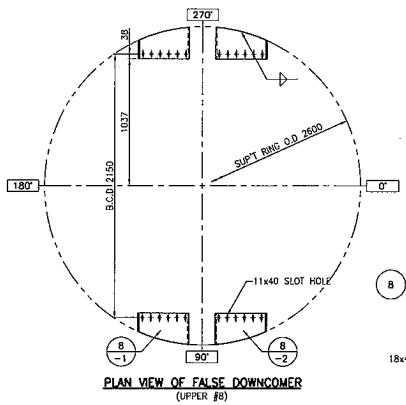
TITLE:
103-C-101/201
CONDENSATE STABILISER
TOWER ATTACHMENT DWG.(2/3)

DWG No. VP-6340S-103-C1101-005 | SHEET No. 3 OF 11 | REV. F

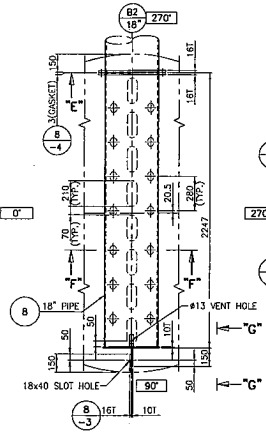
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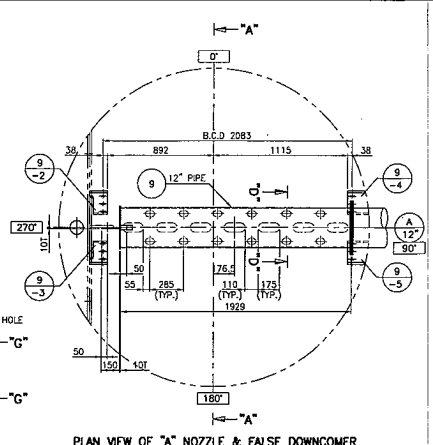
PLAN VIEW OF CHIMNEY TRAY SUPPORT RING
(BELOW #2)



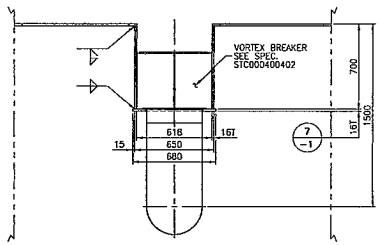
PLAN VIEW OF FALSE DOWNCOMER
(UPPER #8)



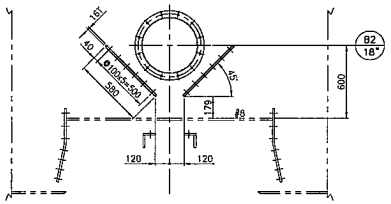
PLAN VIEW OF "B2" NOZZLE
(UPPER #8)



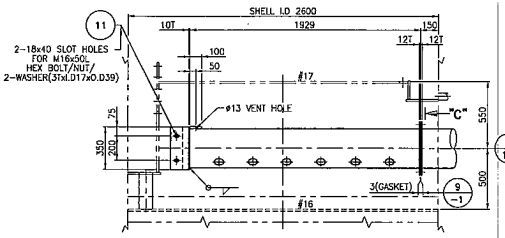
PLAN VIEW OF "A" NOZZLE & FALSE DOWNCOMER
(UPPER #16)



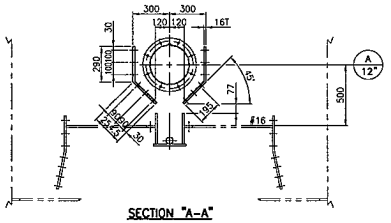
SECTION THRU CENTER LINE FOR CHIMNEY TRAY SUPPORT



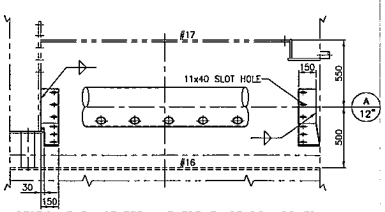
SECTION THRU CENTER LINE FOR FALSE DOWNCOMER



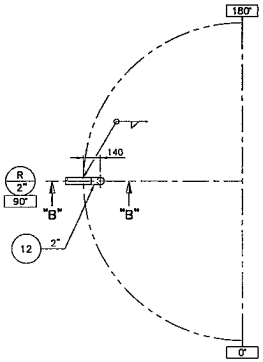
SECTION THRU CENTER LINE FOR "A" NOZZLE



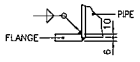
SECTION "A-A"



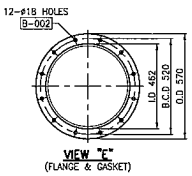
SECTION THRU CENTER LINE FOR FALSE DOWNCOMER



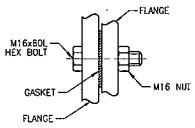
PLAN VIEW OF "R" NOZZLE
(UPPER #19)



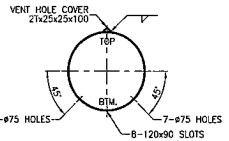
TYPICAL WELD DETAIL
(FOR "B2" NOZZLE)



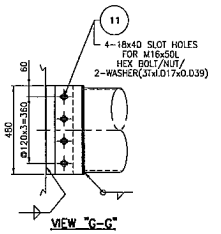
VIEW "C"
(FLANGE & GASKET)



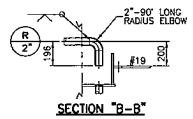
DETAIL "B-002" 13



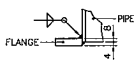
VIEW "F-F"



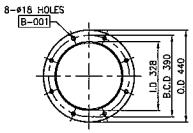
VIEW "G-G"



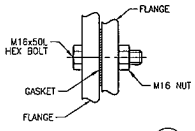
SECTION "B-B"



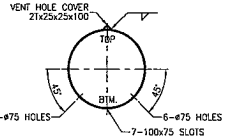
TYPICAL WELD DETAIL
(FOR "A" NOZZLE)



VIEW "C"
(FLANGE & GASKET)



DETAIL "B-001" 10



VIEW "D-D"

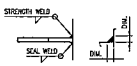
NO.		DESCRIPTION	MAT'L	Q'TY NET	Q'TY UNIT	REMARKS
7		SUPPORT RING	A516GR60N	1	SET	16Tx65W
	-1	SUMP	A516GR60N	1	EA	16T
8		FEED PIPE	A312-TP304L	1	SET	18" PIPE/16T/10T
	-1	FALSE DOWNCOMER	A516GR60N	2	EA	16T
	-2	FALSE DOWNCOMER	A516GR60N	2	EA	16T
	-3	CUP	A516GR60N	1	EA	16T
	-4	GSAKET	VALQUA-#6500	1	EA	3T(NON-ASBESTOS)
9		FEED PIPE	A312-TP304L	1	SET	12" PIPE/12T/10T
	-1	GSAKET	VALQUA-#6500	1	EA	3T(NON-ASBESTOS)
	-2	FALSE DOWNCOMER	A516GR60N	1	EA	16T
	-3	FALSE DOWNCOMER	A516GR60N	1	EA	16T
	-4	FALSE DOWNCOMER	A516GR60N	1	EA	16T
	-5	FALSE DOWNCOMER	A516GR60N	1	EA	16T
10		HEX BOLT/NUT	A193-B8/A194-8	8	SETS	M16x50L
11		HEX BOLT/NUT/2-WASHER	A193-B8/A194-8 A36-SS1	6	SETS	M16x50L/ 3T
12		FEED PIPE	A105-B	1	EA	12"
13		HEX BOLT/NUT	A193-B8/A194-8	12	SETS	M16x60L

(TOTAL; 2 SETS)

GENERAL INFORMATION

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETERS.
2. ALL TOWER ATTACHMENTS SHOWN ON THIS DRAWING BE NOT ACTUAL FABRICATION DRAWING. THEREFORE, THE FABRICATION DRAWING SHOULD BE PREPARED BY VESSEL VENDOR.
3. ALL PARTS SHOWN ON THIS DRAWING SHALL BE SUPPLIED AND INSTALLED BY VESSEL FABRICATOR.
4. ALL TRAY ELEVATIONS ARE TO TOP OF THE TRAY SUPPORT RINGS.
5. SUPPORT RINGS ARE TO BE LEVEL ALL AROUND CIRCUMFERENCE OF TOWER, ACROSS TOWER, ACROSS SUPPORT RING FROM TOWER TO INNER EDGE.
6. GRIND OFF ALL BURRS AND SLUG ON THE TOP SURFACE OF SUPPROT RING.
7. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013.

TYPICAL WELD DETAILS



NOTE : All welds are continuous 11mm fillet welds unless otherwise specified.

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
2	23/06/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
1	27/04/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
0	05/03/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

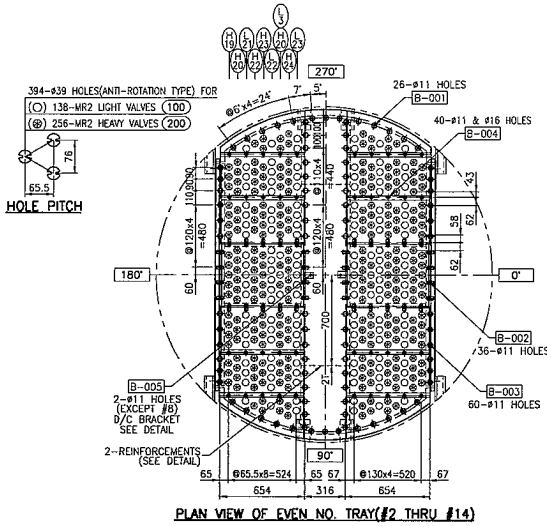
PROJECT No.: POGC-760-81-156 | DOC. CLASS. | SCALE : NONE

HMT | OIEC | TOPC

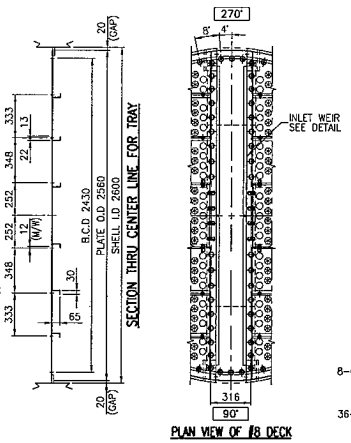
TITLE:
103-C-101/201
CONDENSATE STABILISER
TOWER ATTACHMENT DWG.(3/3)

DWG No. VP-6340S-103-C1101-005 | SHEET No. 4 OF 11 | REV. F

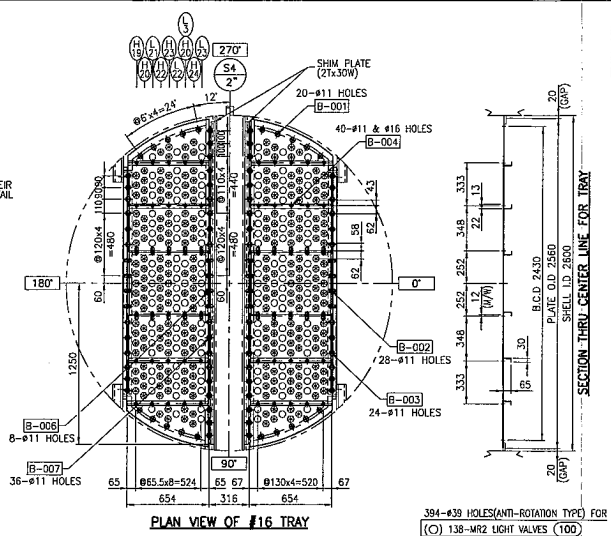
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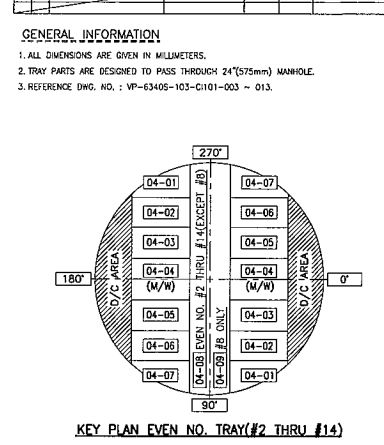
PLAN VIEW OF EVEN NO. TRAY(#2 THRU #14)



PLAN VIEW OF #8 DECK



PLAN VIEW OF #16 TRAY

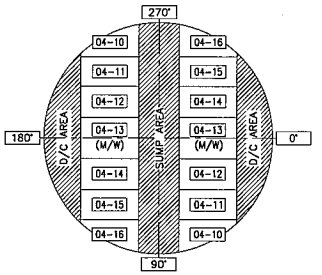


NO.	DESCRIPTION	MAT'L	Q'TY NET	Q'TY UNIT	REMARKS
					VP-6340S-103-C1101-012, 013.

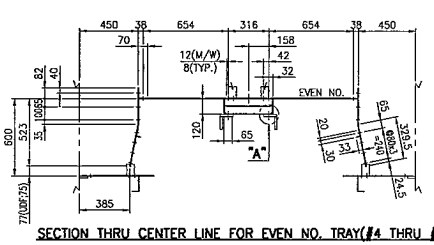
GENERAL INFORMATION

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETERS.
2. TRAY PARTS ARE DESIGNED TO PASS THROUGH 24"(575mm) MANHOLE.
3. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013.

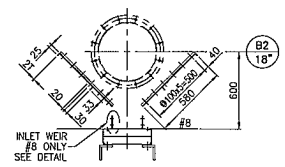
KEY PLAN EVEN NO. TRAY(#2 THRU #14)



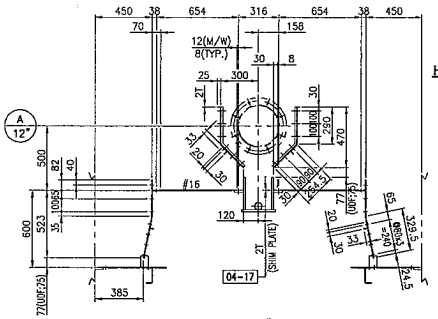
KEY PLAN #16 TRAY



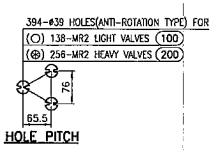
SECTION THRU CENTER LINE FOR EVEN NO. TRAY(#4 THRU #14)

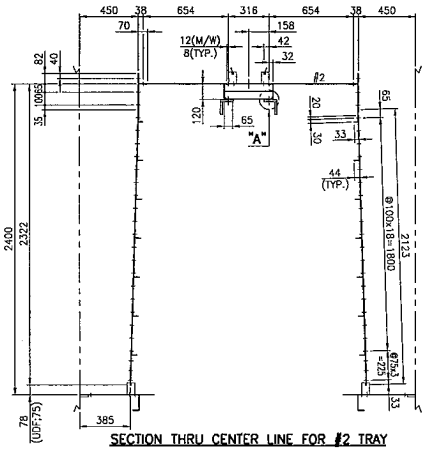


SECTION THRU CENTER LINE FOR FALSE DOWNCOMER

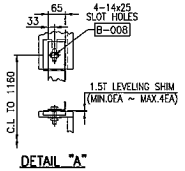


SECTION THRU CENTER LINE FOR #16 TRAY & FALSE DOWNCOMER

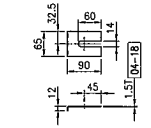




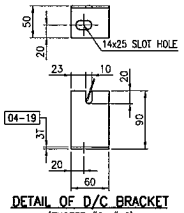
SECTION THRU CENTER LINE FOR #2 TRAY



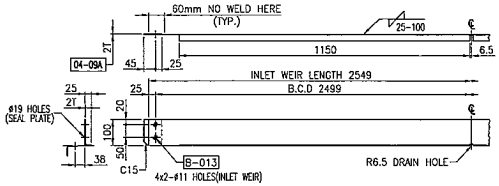
DETAIL "A"



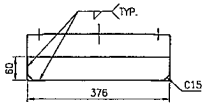
DETAIL OF LEVELING SHIM
(FOR EVEN NO. #2 THRU #14)



DETAIL OF D/C BRACKET
(EXCEPT #8, #16)



DETAIL OF INLET WEIR
(#8 ONLY)



DETAIL OF REINFORCEMENTS

THIS DRAWING CONTAINS CONFIDENTIAL AND PROPRIETARY INFORMATION AND SHALL NOT BE COPIED OR OTHERWISE DIVULGED IN WHOLE OR IN PART WITHOUT THE EXPRESS WRITTEN CONSENT OF HANBAL MASSTECH LIMITED.

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
1	23/07/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
0	13/06/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

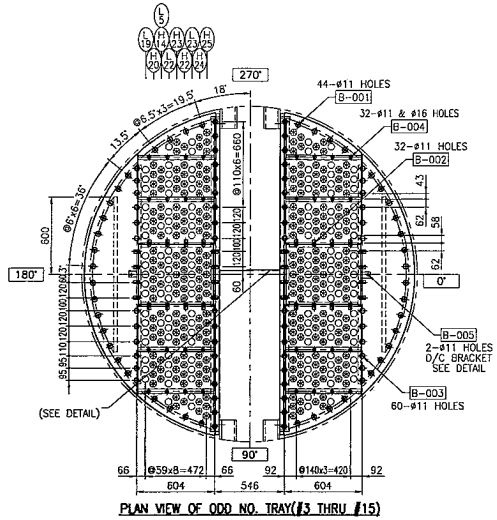
N.I.O.C
PARS OIL AND GAS COMPANY

PROJECT No.: POGC-760-81-156	DOC. CLASS.	SCALE : NONE

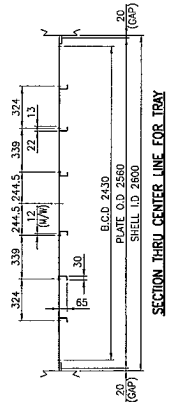
HMT | OIEC | POGC

TITLE:
103-C-101/201
CONDENSATE STABILISER
TRAY DETAIL DRAWING (1/5)

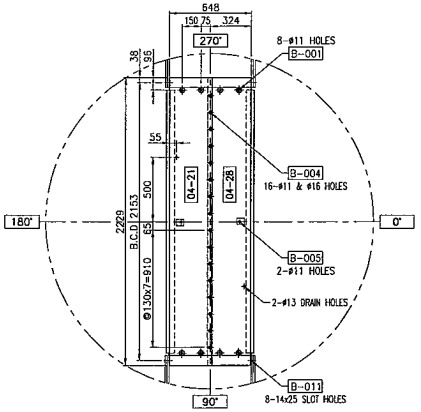
DWG No.	VP-6340S-103-C1101-007	SHEET No. 5 OF 11	REV. F



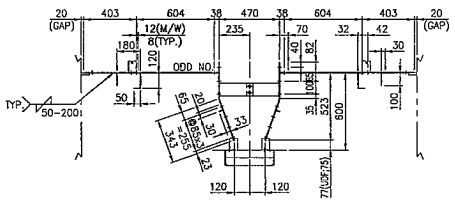
PLAN VIEW OF ODD NO. TRAY(#3 THRU #15)



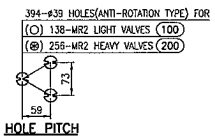
SECTION THRU CENTER LINE FOR TRAY



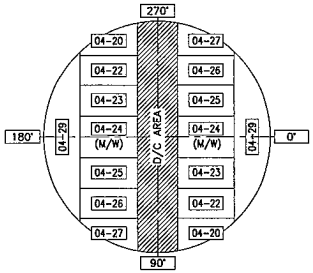
PLAN VIEW OF SEAL PAN
(BELOW #9)



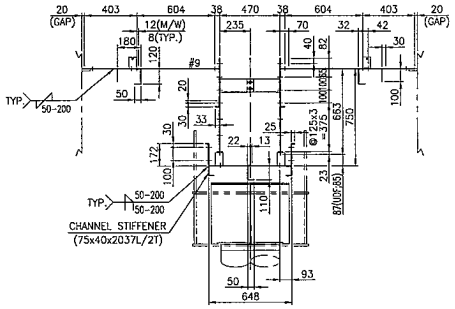
SECTION THRU CENTER LINE FOR ODD NO. TRAY(#3 THRU #15)
(EXCEPT #9)



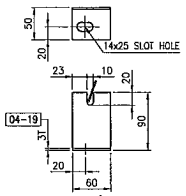
HOLE PITCH



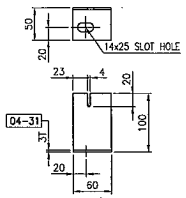
KEY PLAN ODD NO. TRAY(#3 THRU #15)



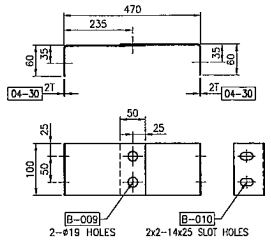
SECTION THRU CENTER LINE FOR #9 TRAY
& SEAL PAN



DETAIL OF D/C BRACKET



DETAIL OF D/C BRACKET
(FOR SEAL PAN)



DETAIL OF DOWNCOMER BRACES

GENERAL INFORMATION

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETERS.
2. TRAY PARTS ARE DESIGNED TO PASS THROUGH 24"(575mm) MANHOLE.
3. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013.

NO.	DESCRIPTION	MAT'L	Q'TY NET	Q'TY UNIT	REMARKS
					VP-6340S-103-C1101-012, 013.

FINAL
2006. 6. 19
HANBAL MASSTECH
LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
1	23/07/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
0	13/06/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

PROJECT No.: POGC-760-81-156	DOC. CLASS.	SCALE : NONE

HMT | OIEC | IOEC

TITLE:
103-C-101/201
CONDENSATE STABILISER
TRAY DETAIL DRAWING (2/5)

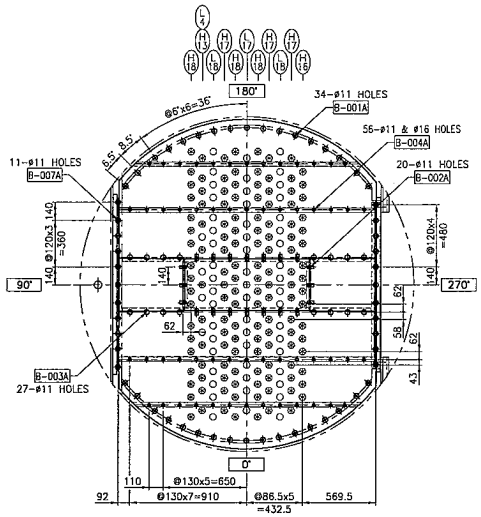
DWG No.	VP-6340S-103-C1101-008	SHEET No. 8 OF 11	REV. F

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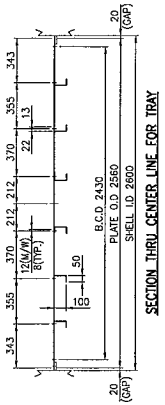
NO.	DESCRIPTION	MAT'L	Q'TY NET	Q'TY UNIT	REMARKS
					VP-6340S-103-C1101-012, 013.

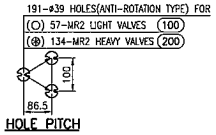
GENERAL INFORMATION

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETERS.
2. TRAY PARTS ARE DESIGNED TO PASS THROUGH 24"(575mm) MANHOLE.
3. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013.

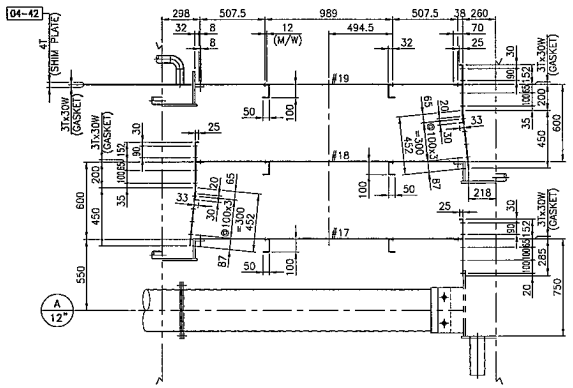


PLAN VIEW OF #17, #19 TRAY
ROTATED 180° FOR #18 TRAY

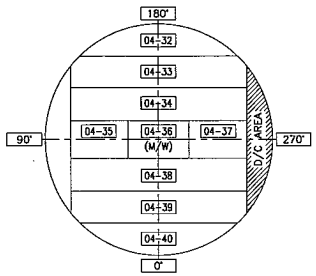




HOLE PITCH



SECTION THRU CENTER LINE FOR #17, #18, #19 TRAY



KEY PLAN #17, #19 TRAY
ROTATED 180° FOR #18 TRAY

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
1	23/07/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
0	13/06/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

PROJECT No.: POGC-760-81-156	DOC. CLASS.	SCALE : NONE

HMT | OIEC | I.O.E.C

THIS DRAWING CONTAINS CONFIDENTIAL AND PROPRIETARY INFORMATION AND SHALL NOT BE COPIED OR OTHERWISE DIVULGED IN WHOLE OR IN PART WITHOUT THE EXPRESS WRITTEN CONSENT OF HANBAL MASSTECH LIMITED.

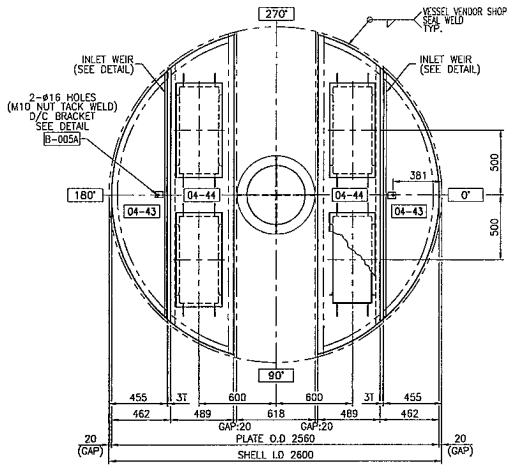
TITLE:
103-C-101/201
CONDENSATE STABILISER
TRAY DETAIL DRAWING (3/5)

DWG No.	VP-6340S-103-C1101-009	SHEET No. 7 OF 11	REV. F

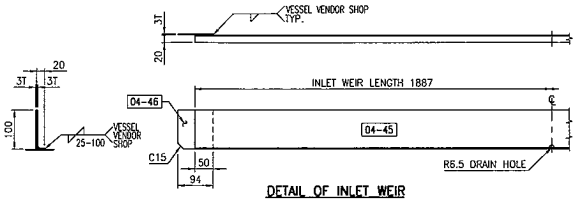
NO.	DESCRIPTION	MAT'L	Q'TY NET	Q'TY UNIT	REMARKS
					VP-6340S-103-C1101-012, 013.

GENERAL INFORMATION

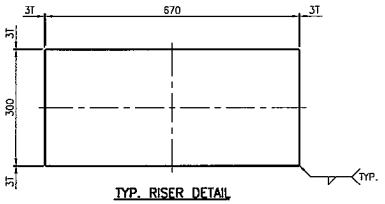
1. ALL DIMENSIONS ARE GIVEN IN MILLIMETERS.
2. TRAY PARTS ARE DESIGNED TO PASS THROUGH 24"(575mm) MANHOLE.
3. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013.
4. CHIMNEY TRAY ARE FURNISHED BY HMT. INSTALLED BY VESSEL VENDOR.



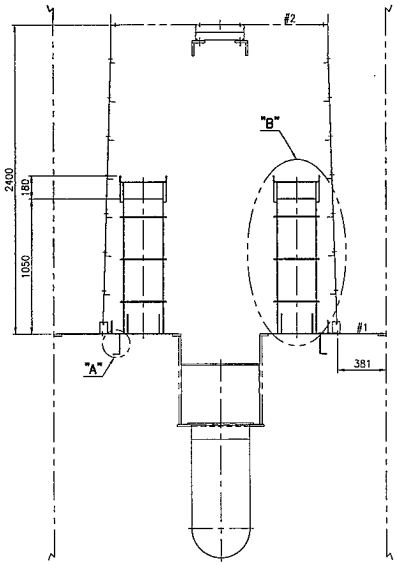
PLAN VIEW OF CHIMNEY TRAY
(BELOW #2)



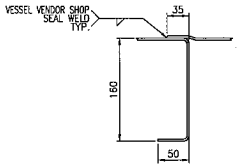
DETAIL OF INLET WEIR



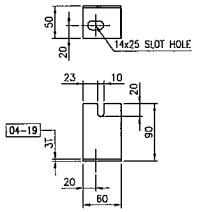
TYP. RISER DETAIL



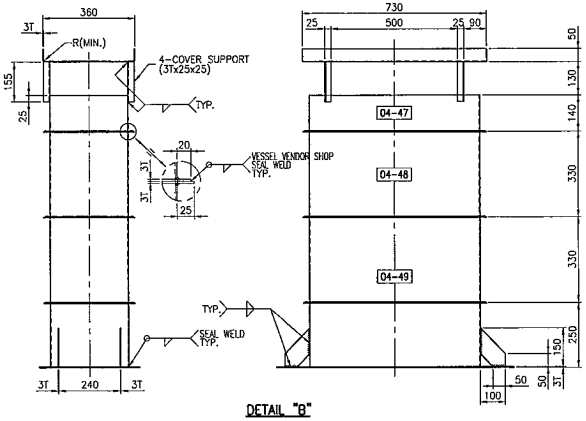
SECTION THRU CENTER LINE FOR CHIMNEY TRAY



DETAIL "A"



DETAIL OF D/C BRACKET



DETAIL "B"

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
1	23/07/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
0	13/06/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

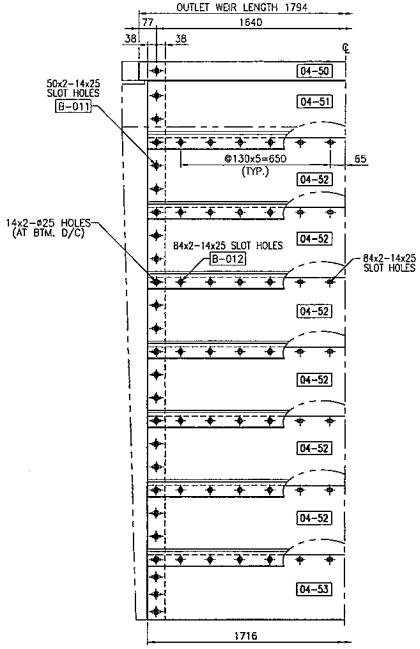
PROJECT No.: POGC-760-81-156 | DOC. CLASS. | SCALE : NONE

HMT | OIEC | IOEC

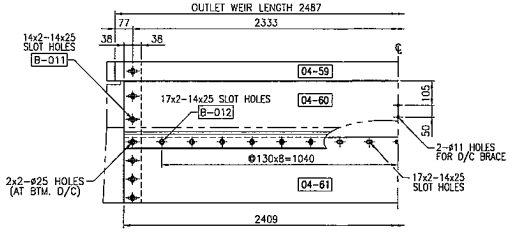
TITLE: 103-C-101/201
CONDENSATE STABILISER
TRAY DETAIL DRAWING (4/5)

DWG No. VP-6340S-103-C1101-010 | SHEET No. 8 OF 11 | REV. F

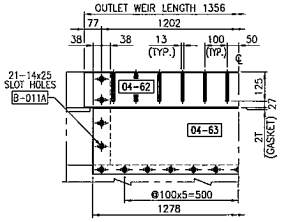
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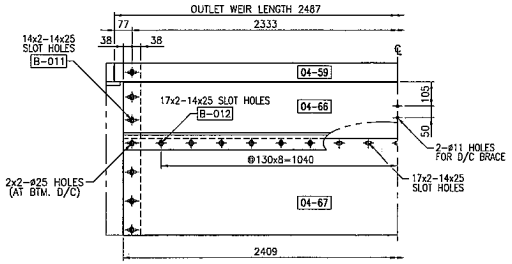
DETAIL OF #2 SIDE DOWNCOMER



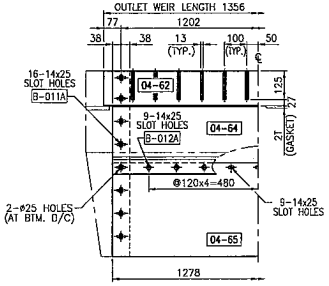
DETAIL OF CENTER DOWNCOMER(#3 THRU #15)
(EXCEPT #9)



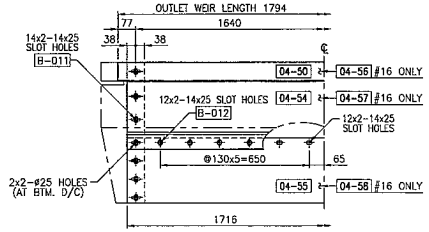
DETAIL OF #17 DOWNCOMER



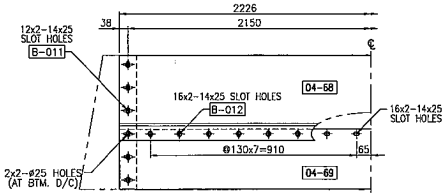
DETAIL OF CENTER DOWNCOMER(#9 ONLY)



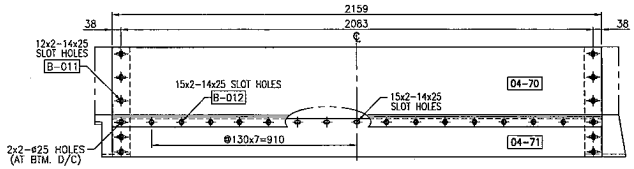
DETAIL OF #18, #19 DOWNCOMER



DETAIL OF SIDE DOWNCOMER(#4 THRU #16)



DETAIL OF FALSE DOWNCOMER
(ABOVE #8)



DETAIL OF FALSE DOWNCOMER
(ABOVE #16)

NO.	DESCRIPTION	MAT'L	QTY NET	QTY UNIT	REMARKS
					VP-6340S-103-C1101-012, 013.

GENERAL INFORMATION

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETERS.
2. TRAY PARTS ARE DESIGNED TO PASS THROUGH 24"(575mm) MANHOLE.
3. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013.

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
1	23/07/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
0	13/06/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

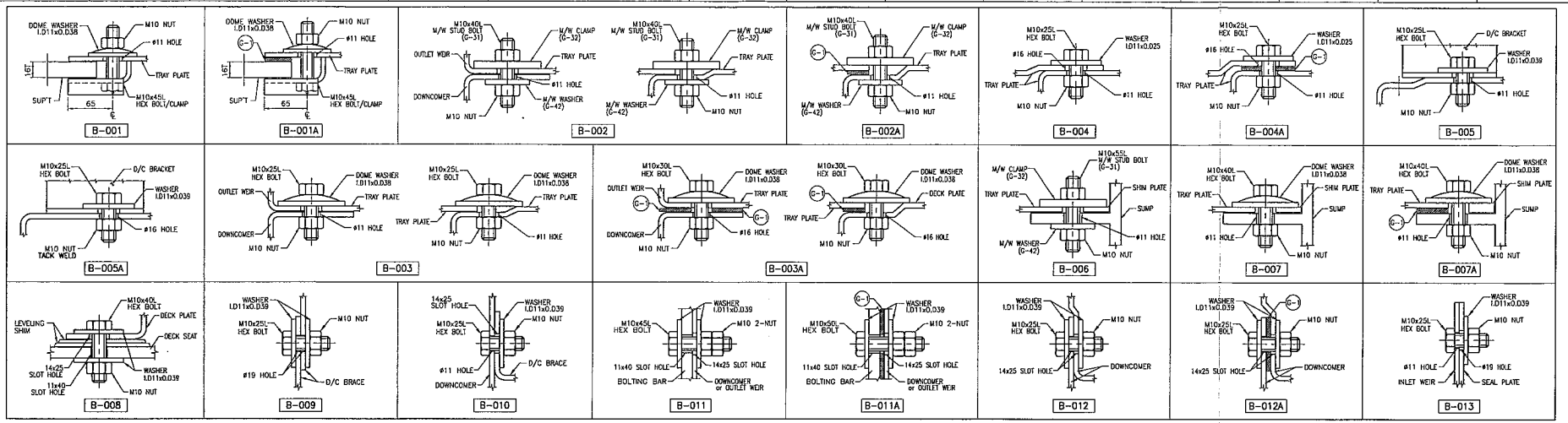
PROJECT No.: POGC-760-81-156 | DOC. CLASS. | SCALE : NONE

HMT | OIEC | IOEC

TITLE:
103-C-101/201
CONDENSATE STABILISER
TRAY DETAIL DRAWING (5/5)

DWG No. VP-6340S-103-C1101-011 | SHEET No. 9 OF 11 | REV. F

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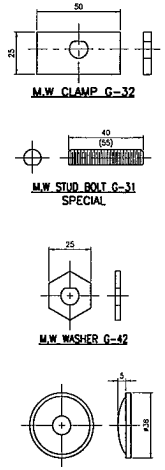


PARTS LIST

PART NO.	DESCRIPTION	MATERIAL	SIZE	NET Q'TY	SPARES (CON. & COMM.)	SPARES (2 YEARS)	TOTAL Q'TY	Q'TY PER INTERNAL: EVEN NO. TRAY (#2 ONLY) (1)	EVEN NO. TRAY (#4 THRU #14) (EXCEPT #6) (5)	EVEN NO. TRAY (#6 ONLY) (1)	EVEN NO. TRAY (#16 ONLY) (1)	ODD NO. TRAY (#3 THRU #15) (EXCEPT #9) (6)	ODD NO. TRAY (#9 ONLY) (1)	SEAL PAN (BELOW #9) (1)	TRAY (#17 ONLY) (1)	TRAY (#18, #19) (2)	CHIMNEY TRAY (BELOW #2) (1)
G-1	GASKET	FIBER-GLASS	31x30W	73.8M	18.5M	–	92.3M								24.4M	24.7M	
G-31	M/W STUD BOLT	A193-B8	M10x40L	476	48	–	524	36	36	36		32	32				
G-31	M/W STUD BOLT	A193-B8	M10x55L	8	10	–	18						8				
(B-001)	HEX BOLT/CLAMP	A193-B8/A240-304L	M10x45L/2T	498	50	–	548	26	26	26		44	44	8			
	HEX BOLT	A193-B8	M10x45L	496	50	–	546	100	28	52		28	28	8			
	HEX BOLT	A193-B8	M10x40L	64	10	–	74	4	4	4			36				
	HEX BOLT	A193-B8	M10x25L	2022	203	–	2225	270	126	164		134	134	18			2
	HEX NUT	A194-8	M10 NUT	4542	455	–	4997	572	284	370		298	350	42			
	HEX NUT(TACK WELD)	A194-8	M10 NUT	2	–	–	2										2
G-32	M/W CLAMP	A240-304L	5T	484	49	–	533	36	36	36		32	40				
G-42	M/W WASHER	A240-304L	2T	484	49	–	533	36	36	36		32	40				
K-74452	DOME WASHER	A240-304L	2Tx1.D11x0.D38	1374	138	–	1512	86	86	86		104	140	8			
	FLAT WASHER	A240-304L	2Tx1.D11x0.D39	2306	231	–	2537	546	114	232		134	134	18			2
	FLAT WASHER	A240-304L	2Tx1.D11x0.D25	520	52	–	572	40	40	40		32	32	16			
100	LIGHT VALVE	A240-304L	MR2	1932	97	–	2029	138	138	138		138	138				
200	HEAVY VALVE	A240-304L	MR2	3584	180	–	3764	256	256	256		256	256				
G-31	M/W STUD BOLT	A193-B8T	M10x40L	88	10	–	98				28				20	20	
G-31	M/W STUD BOLT	A193-B8T	M10x55L	8	10	–	18				8						
(B-001)	HEX BOLT/CLAMP	A193-B8T/A240-321	M10x45L/2T	20	5	–	25				20						
(B-001A)	HEX BOLT/CLAMP	A193-B8T/A240-321	M10x45L/2T	102	11	–	113								34	34	
	HEX BOLT	A193-B8T	M10x50L	53	10	–	63								21	16	
	HEX BOLT	A193-B8T	M10x45L	52	10	–	62				52						
	HEX BOLT	A193-B8T	M10x40L	69	10	–	79				36				11	11	
	HEX BOLT	A193-B8T	M10x30L	81	10	–	91								27	27	
	HEX BOLT	A193-B8T	M10x25L	304	31	–	335				118				56	65	
	HEX NUT	A194-8T	M10 NUT	978	98	–	1076				350				210	209	
G-32	M/W CLAMP	A240-321	5T	96	10	–	106				36				20	20	
G-42	M/W WASHER	A240-321	2T	96	10	–	106				36				20	20	
K-74452	DOME WASHER	A240-321	2Tx1.D11x0.D38	296	30	–	326				80				72	72	
	FLAT WASHER	A240-321	2Tx1.D11x0.D39	354	36	–	390				212				42	50	
	FLAT WASHER	A240-321	2Tx1.D11x0.D25	208	21	–	229				40				56	56	
100	LIGHT VALVE	A240-321	MR2	309	16	–	325				138				57	57	
200	HEAVY VALVE	A240-321	MR2	658	33	–	691				256				134	134	

ASSEMBLIES PER INTERNAL

ASSEMBLY NUMBER	EVEN NO. TRAY (#2 ONLY) (1)	EVEN NO. TRAY (#4 THRU #14) (EXCEPT #6) (5)	EVEN NO. TRAY (#6 ONLY) (1)	EVEN NO. TRAY (#16 ONLY) (1)	ODD NO. TRAY (#3 THRU #15) (EXCEPT #9) (6)	ODD NO. TRAY (#9 ONLY) (1)	SEAL PAN (BELOW #9) (1)	TRAY (#17 ONLY) (1)	TRAY (#18, #19) (2)	CHIMNEY TRAY (BELOW #2) (1)	TOTAL
B-001	26	26	26	20	44	44	8				518
B-001A								34	34		102
B-002	36	36	36	28	32	32					504
B-002A								20	20		60
B-003	60	60	60	24	60	60					864
B-003A								27	27		81
B-004	40	40	40	40	32	32	16				560
B-004A								56	56		168
B-005	2	2			2	2	2				28
B-005A										2	2
B-006				8		8					16
B-007				36		36					72
B-007A								11	11		33
B-008	4	4	4								28
B-009					2	2					14
B-010					4	4					28
B-011	100	28	52	52	28	28	8				548
B-011A								21	16		53
B-012	168	24	56	54	34	34					636
B-012A									9		18
B-013			8								8



FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

GENERAL INFORMATION

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETERS.
2. REFERENCE DWG. NO. : VP-6340S-103-CI101-003 ~ 013.
3. SPARE PARTS
   1) FOR COMMISSIONING / CONSTRUCTION
   - BOLT/NUT/WASHER : 10% (MIN. 10 PCS OF EACH TYPE)
   - CLAMP : 10% (MIN. 5 PCS)
   - GASKET : 25% (10 FEET)
   - VALVE CAPS : 5%

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/05/06	FINAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
1	23/07/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
0	13/06/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

PROJECT No.: PGGC-760-81-156 | DOC. CLASS. | SCALE : NONE

HMT | OIEC | TOBC

TITLE:
103-C-101/201
CONDENSATE STABILISER
BOLTING ASS'Y DWG.

DWG No. VP-6340S-103-CI101-012 | SHEET No. 10 OF 11 | REV. F

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PARTS LIST

PART NO.	DESCRIPTION	MATERIAL	SIZE	NET Q'TY	SPARES (CON. & COMM.)	SPARES (2 YEARS)	TOTAL Q'TY	Q'TY PER INTERNAL										
								EVEN NO. TRAY (#16 ONLY)	EVEN NO. TRAY (#4 THRU #14)	EXCEPT #8)	EVEN NO. TRAY (#8 ONLY)	ODD NO. TRAY (#16 ONLY)	ODD NO. TRAY (#3 THRU #15) EXCEPT #9)	ODD NO. TRAY (#9 ONLY)	SEAL PAN (BELOW #9)	TRAY (#17 ONLY)	TRAY (#18, #19)	CHIMNEY TRAY (BELOW #2)
								(1)	(5)	(1)	(1)	(1)	(6)	(1)	(1)	(1)	(2)	(1)
04-01	TRAY PLATE	A240-304L	2T	14	-	-	14	2	2	2								
04-02	TRAY PLATE	A240-304L	2T	14	-	-	14	2	2	2								
04-03	TRAY PLATE	A240-304L	2T	14	-	-	14	2	2	2								
04-04	TRAY PLATE(M/W)	A240-304L	2T	14	-	-	14	2	2	2								
04-05	TRAY PLATE	A240-304L	2T	14	-	-	14	2	2	2								
04-06	TRAY PLATE	A240-304L	2T	14	-	-	14	2	2	2								
04-07	TRAY PLATE	A240-304L	2T	14	-	-	14	2	2	2								
04-08	CENTER DECK	A240-304L	2T	6	-	-	6	1	1									
04-09	CENTER DECK	A240-304L	2T	1	-	-	1				1							
04-09A	SEAL PLATE	A240-304L	2T	4	-	-	4				4							
04-10	TRAY PLATE	A240-321	2T	2	-	-	2					2						
04-11	TRAY PLATE	A240-321	2T	2	-	-	2					2						
04-12	TRAY PLATE	A240-321	2T	2	-	-	2					2						
04-13	TRAY PLATE(M/W)	A240-321	2T	2	-	-	2					2						
04-14	TRAY PLATE	A240-321	2T	2	-	-	2					2						
04-15	TRAY PLATE	A240-321	2T	2	-	-	2					2						
04-16	TRAY PLATE	A240-321	2T	2	-	-	2					2						
04-17	SHIM PLATE	A240-321	2T	2	-	-	2					2						
04-18	LEVELING SHIM	A240-304L	1.5T	112	-	-	112	16	16	16	16							
04-19	DOWNCOMER BRACKET	A240-304L	3T	28	-	-	28	2	2				2	2				2
04-20	TRAY PLATE	A240-304L	2T	14	-	-	14						2	2				
04-21	SEAL PAN	A240-304L	2T	1	-	-	1								1			
04-22	TRAY PLATE	A240-304L	2T	14	-	-	14						2	2				
04-23	TRAY PLATE	A240-304L	2T	14	-	-	14						2	2				
04-24	TRAY PLATE(M/W)	A240-304L	2T	14	-	-	14						2	2				
04-25	TRAY PLATE	A240-304L	2T	14	-	-	14						2	2				
04-26	TRAY PLATE	A240-304L	2T	14	-	-	14						2	2				
04-27	TRAY PLATE	A240-304L	2T	14	-	-	14						2	2				
04-28	SEAL PAN	A240-304L	2T	1	-	-	1								1			
04-29	SIDE DECK	A240-304L	2T	14	-	-	14						2	2				
04-30	DOWNCOMER BRACE	A240-304L	2T	14	-	-	14						2	2				
04-31	DOWNCOMER BRACKET	A240-304L	3T	2	-	-	2								2			
04-32	TRAY PLATE	A240-321	2T	3	-	-	3									1	1	
04-33	TRAY PLATE	A240-321	2T	3	-	-	3									1	1	
04-34	TRAY PLATE	A240-321	2T	3	-	-	3									1	1	
04-35	TRAY PLATE	A240-321	2T	3	-	-	3									1	1	
04-36	TRAY PLATE(M/W)	A240-321	2T	3	-	-	3									1	1	
04-37	TRAY PLATE	A240-321	2T	3	-	-	3									1	1	
04-38	TRAY PLATE	A240-321	2T	3	-	-	3									1	1	
04-39	TRAY PLATE	A240-321	2T	3	-	-	3									1	1	
04-40	TRAY PLATE	A240-321	2T	3	-	-	3									1	1	
04-42	SHIM PLATE	A240-321	4T	3	-	-	3									1	1	
04-43	CHIMNEY TRAY SECTION	A240-304L	3T	2	-	-	2											2
04-44	CHIMNEY TRAY SECTION	A240-304L	3T	2	-	-	2											2
04-45	INLET WEIR	A240-304L	3T	2	-	-	2											2
04-46	SEAL PLATE	A240-304L	3T	4	-	-	4											4
04-47	RISER SECTION	A240-304L	3T	4	-	-	4											4
04-48	RISER SECTION	A240-304L	3T	4	-	-	4											4
04-49	RISER SECTION	A240-304L	3T	4	-	-	4											4
04-50	OUTLET WEIR	A240-304L	2T	14	-	-	14	2	2	2								
04-51	DOWNCOMER	A240-304L	2T	2	-	-	2	2										
04-52	DOWNCOMER	A240-304L	2T	12	-	-	12	12										
04-53	DOWNCOMER	A240-304L	2T	2	-	-	2	2										
04-54	DOWNCOMER	A240-304L	2T	12	-	-	12		2	2								
04-55	DOWNCOMER	A240-304L	2T	12	-	-	12		2	2								
04-56	OUTLET WEIR	A240-321	2T	2	-	-	2					2						
04-57	DOWNCOMER	A240-321	2T	2	-	-	2					2						
04-58	DOWNCOMER	A240-321	2T	2	-	-	2					2						
04-59	OUTLET WEIR	A240-304L	2T	14	-	-	14						2	2				
04-60	DOWNCOMER	A240-304L	2T	12	-	-	12						2					
04-61	DOWNCOMER	A240-304L	2T	12	-	-	12						2					
04-62	OUTLET WEIR	A240-321	2T	3	-	-	3									1	1	
04-63	DOWNCOMER	A240-321	2T	1	-	-	1									1		
04-64	DOWNCOMER	A240-321	2T	2	-	-	2										1	
04-65	DOWNCOMER	A240-321	2T	2	-	-	2										1	
04-66	DOWNCOMER	A240-304L	2T	2	-	-	2							2				
04-67	DOWNCOMER	A240-304L	2T	2	-	-	2							2				
04-68	FALSE DOWNCOMER	A240-304L	2T	2	-	-	2				2							
04-69	FALSE DOWNCOMER	A240-304L	2T	2	-	-	2				2							
04-70	FALSE DOWNCOMER	A240-321	2T	2	-	-	2					2						
04-71	FALSE DOWNCOMER	A240-321	2T	2	-	-	2					2						

PARTS LIST

PART NO.	DESCRIPTION	MATERIAL	SIZE	NET Q'TY	SPARES (CON. & COMM.)	SPARES (2 YEARS)	TOTAL Q'TY	Q'TY PER INTERNAL

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

GENERAL INFORMATION

1. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
1	23/07/05	CONFIRMED THE COMMENTS	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
0	13/06/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

PROJECT No.: POGC-760-81-156 | DOC. CLASS. | SCALE : NONE

HMT | OIEC | IOEC

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TITLE:
103-C-101/201
CONDENSATE STABILISER
PART LIST DWG.

DWG No.	SHEET No. 11 OF 11	REV. F
VP-6340S-103-C1101-013		